

fong-spivak-seven-sketches: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-EQ0061 (3.61) [conf 0.008]	110	■ 0.008	$I = \left[\begin{array}{ccc} 1 & \underset{\bullet}{a} & \underset{\bullet}{2} \\ & \downarrow & \downarrow \\ & \downarrow & \downarrow \\ & \downarrow & \bullet \end{array} \right] b d$	$I = \left[\begin{array}{ccc} 1 & \underset{\bullet}{a} & \underset{\bullet}{2} \\ & \downarrow & \downarrow \\ & \downarrow & \downarrow \\ & \downarrow & \bullet \end{array} \right] b d$	$I = \begin{array}{c} \begin{array}{ccc} 1 & \xrightarrow{a} & 2 \\ & \searrow b & \downarrow e \\ & & 3 \end{array} \quad \begin{array}{c} \curvearrowright d \\ \bullet \\ 4 \end{array} \end{array}$
fong-spivak-seven-sketches-EQ0063 [conf 0.014]	111	■ 0.014	$G := \stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b} \stackrel{3}{\bullet} \quad H := \stackrel{4}{\bullet} \xrightarrow[c]{d} \stackrel{5}{\bullet} \curvearrowright e \quad \text{Pe}$	$G := \stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b} \stackrel{3}{\bullet} \quad H := \stackrel{4}{\bullet} \xrightarrow[c]{d} \stackrel{5}{\bullet} \curvearrowright e \quad \text{Pe}$	$G := \begin{array}{ccc} 1 & \xrightarrow{a} & 2 \\ & \searrow b & \downarrow e \\ & & 3 \end{array} \quad \begin{array}{c} \curvearrowright d \\ \bullet \\ 4 \end{array}$
fong-spivak-seven-sketches-EQ0133 [conf 0.025]	214	■ 0.025	$\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{[\text{id}_X, \text{id}_X]} X \xleftarrow{\text{id}_X} X \right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\text{id}_X} X \right) \\ \delta_X &:= \left(X \xrightarrow{\text{id}_X} X \xleftarrow{[\text{id}_X, \text{id}_X]} X + X \right) \\ \epsilon_X &:= \left(X \xrightarrow{\text{id}_X} X \xleftarrow{!_X} \emptyset \right) \end{aligned}$	$\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{[\text{id}_X, \text{id}_X]} X \xleftarrow{\text{id}_X} X \right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\text{id}_X} X \right) \\ \delta_X &:= \left(X \xrightarrow{\text{id}_X} X \xleftarrow{[\text{id}_X, \text{id}_X]} X + X \right) \\ \epsilon_X &:= \left(X \xrightarrow{\text{id}_X} X \xleftarrow{!_X} \emptyset \right) \end{aligned}$	$\begin{aligned} \mu_X &:= (X + X \xrightarrow{[\text{id}_X, \text{id}_X]} X \xleftarrow{\text{id}_X} X) \\ \eta_X &:= (\emptyset \xrightarrow{!_X} X \xleftarrow{\text{id}_X} X) \\ \delta_X &:= (X \xrightarrow{\text{id}_X} X \xleftarrow{[\text{id}_X, \text{id}_X]} X + X) \\ \epsilon_X &:= (X \xrightarrow{\text{id}_X} X \xleftarrow{!_X} \emptyset) \end{aligned}$
fong-spivak-seven-sketches-EQ0188 [conf 0.038]	279	■ 0.038	$g_!(c_3) = \{ \stackrel{12}{\downarrow} \stackrel{3}{\downarrow} \stackrel{4}{\downarrow} \}$	$g_!(c_3) = \stackrel{12}{\downarrow} \stackrel{3}{\downarrow} \stackrel{4}{\downarrow}$	$g_!(c_3) = \begin{array}{ccc} 12 & 3 & 4 \\ \downarrow & \downarrow & \downarrow \\ \bullet & & \bullet \end{array}$
fong-spivak-seven-sketches-EQ0181 [conf 0.044]	274	■ 0.044	$G = \left[\begin{array}{ccc} 1 & \underset{\bullet}{a} & \underset{\bullet}{2} \\ & \downarrow & \downarrow \\ & \downarrow & \downarrow \\ & \downarrow & \downarrow \\ & \downarrow & \downarrow \end{array} \right]$	$G = \left[\begin{array}{ccc} 1 & \underset{\bullet}{a} & \underset{\bullet}{2} \\ & \downarrow & \downarrow \\ & \downarrow & \downarrow \\ & \downarrow & \downarrow \\ & \downarrow & \downarrow \end{array} \right]$	$G = \begin{array}{c} \begin{array}{ccc} 1 & \xrightarrow{a} & 2 \\ & \searrow b & \downarrow e \\ & & 3 \end{array} \quad \begin{array}{c} \curvearrowright d \\ \bullet \\ 4 \end{array} \end{array}$
fong-spivak-seven-sketches-EQ0186 [conf 0.050]	279	■ 0.050	$g_!(c_1) = \{ \stackrel{12}{\downarrow} \stackrel{3}{\downarrow} \stackrel{4}{\downarrow} \}$	$g_!(c_1) = \stackrel{12}{\downarrow} \stackrel{3}{\downarrow} \stackrel{4}{\downarrow}$	$g_!(c_1) = \begin{array}{ccc} 12 & 3 & 4 \\ \downarrow & \downarrow & \downarrow \\ \bullet & & \bullet \end{array}$
fong-spivak-seven-sketches-EQ0189	279	■ 0.108	$g_!(c_4) = \{ \stackrel{12}{\downarrow} \stackrel{3}{\downarrow} \stackrel{4}{\downarrow} \}$	$g_!(c_4) = \stackrel{12}{\downarrow} \stackrel{3}{\downarrow} \stackrel{4}{\downarrow}$	$g_!(c_4) = \begin{array}{ccc} 12 & 3 & 4 \\ \downarrow & \downarrow & \downarrow \\ \bullet & & \bullet \end{array}$

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fong-spivak-seven-sketches_1 EQ0073 (3.77)	118	■ 0.114	$\mathcal{G}:=\left \stackrel{\text{Email}}{\underset{\text{received_by}}{\stackrel{\text{sent_by}}{\rightarrow}}}\text{Address}\right ^{\text{no equations}}$	(not rendered)	
fong-spivak-seven-sketches_1 EQ0245	311	■ 0.129	$a_{\mathfrak{m}}:=(1+\cdots+1+\cdots);\cdots;(1+1+\supset-);\cdots:m\rightarrow 1$	$a_m:=(1+\cdots+1+\supset-);\cdots;(1+1+\supset-);\cdots:m\rightarrow 1$	$a_m:=(1+\cdots+1+\supset-);\cdots;(1+1+\supset-);\cdots:m\rightarrow 1$
fong-spivak-seven-sketches_1 EQ0083	136	■ 0.147	$\text{Chassis: }(\text{load}\times\text{velocity})\rightarrow(\text{torque}\times\text{speed}\times\$)$	Chassis: (load × velocity) → (torque × speed × \$)	Chassis: (load × velocity) → (torque × speed × \$)
fong-spivak-seven-sketches_1 EQ0205	290	■ 0.147	$A,\quad A\div f,\quad A\div g,\quad A\div f\div h,\quad A\div g\mathfrak{i}i,\quad B,\quad B\div h,\quad C,\quad C:i,\quad D$	$A,\quad A\div f,\quad A\div g,\quad A\div f\div h,\quad A\div g\mathfrak{i}i,\quad B,\quad B\div h,\quad C,\quad C:i,\quad D$	$A,\quad A\mathbin{\mathcal{A}}f,\quad A\mathbin{\mathcal{A}}g,\quad A\mathbin{\mathcal{A}}f\mathbin{\mathcal{A}}h,\quad A\mathbin{\mathcal{A}}g\mathbin{\mathcal{A}}i,\quad B,\quad B\mathbin{\mathcal{A}}h,\quad C,\quad C\mathbin{\mathcal{A}}i,\quad D$
fong-spivak-seven-sketches_1 EQ0064 (3.65)	112	■ 0.147	$\mathbf{Gr}:=\stackrel{\text{Arrow}}{\rightarrow}\stackrel{\text{source}}{\Rightarrow}\stackrel{\text{Vertex}}{\text{target}}\text{no equations}$	$\mathbf{Gr}:=\stackrel{\text{Arrow}}{\rightarrow}\stackrel{\text{source}}{\Rightarrow}\stackrel{\text{Vertex}}{\text{target}}\text{no equations}$	
fong-spivak-seven-sketches_1 EQ0214	294	■ 0.168	$H:=\stackrel{4}{\rightarrow}\stackrel{d}{\rightarrow}\stackrel{5}{\rightarrow}\bullet>e$	$H:=\stackrel{4}{\rightarrow}\stackrel{d}{\rightarrow}\stackrel{5}{\rightarrow}\bullet>e$	$H:=\stackrel{4}{\rightarrow}\stackrel{d}{\rightarrow}\stackrel{5}{\rightarrow}\bullet\curvearrowright e$
fong-spivak-seven-sketches_1 EQ0115	182	■ 0.203	$G_{\mathcal{R}}:=\{\supset-, \circ-, \subset, \rightarrow\}\cup\{-\mathfrak{a}-\mid a\in R\},$	$G_R:=\{\supset-, \circ-, \subset, \rightarrow\}\cup\{-\mathfrak{a}-\mid a\in R\},$	$G_R:=\left\{\supset-, \circ-, \subset, \rightarrow\right\}\cup\left\{-\mathfrak{a}-\mid a\in R\right\},$
fong-spivak-seven-sketches_1 EQ0187	279	■ 0.215	$g_{\mathfrak{!}}\left(c_{\mathfrak{2}}\right)=\Psi_{\bullet}^{12}\mathfrak{J}^{34^4}$	$g_{\mathfrak{!}}\left(c_2\right)=\Psi_{\bullet}^{12}\mathfrak{J}^{34^4}$	$g_{\mathfrak{!}}\left(c_2\right)=\begin{matrix}12&3&4\\ \downarrow&\downarrow&\downarrow\\ \bullet&\bullet&\bullet\end{matrix}$
fong-spivak-seven-sketches_1 EQ0094	148	■ 0.259	$\begin{array}{l} S\times(T\times U):=\{(s,(t,u))\mid s\in S,t\in T,u\in U\}\\ \mid s\in S,t\in T,u\in U\}\stackrel{?}{=} (S\times T)\times U:=\{((s,t),u)\mid s\in S,t\in T,u\in U\}. \end{array}$	$S\times(T\times U):=\{(s,(t,u))\mid s\in S,t\in T,u\in U\}\stackrel{?}{=} (S\times T)\times U:=\{((s,t),u)\mid s\in S,t\in T,u\in U\}.$	$S\times(T\times U):=\{(s,(t,u))\mid s\in S,t\in T,u\in U\}\stackrel{?}{=} (S\times T)\times U:=\{((s,t),u)\mid s\in S,t\in T,u\in U\}.$
fong-spivak-seven-sketches_1 EQ0072	118	■ 0.263	$\Sigma_{\mathfrak{!}}:\mathcal{C}-\mathcal{I}\backslash\sqcup\rightarrow\mathbf{Set}\quad\text{and}\quad\Pi_{\mathfrak{!}}:\mathcal{C}-\mathbf{Inst}\rightarrow\mathbf{Set}.$	$\Sigma_{\mathfrak{!}}:\mathcal{C}-\mathcal{I}\backslash\sqcup\rightarrow\mathbf{Set}\quad\text{and}\quad\Pi_{\mathfrak{!}}:\mathcal{C}-\mathbf{Inst}\rightarrow\mathbf{Set}.$	$\Sigma_{\mathfrak{!}}:\mathcal{C}\text{-}\mathbf{Inst}\rightarrow\mathbf{Set}\quad\text{and}\quad\Pi_{\mathfrak{!}}:\mathcal{C}\text{-}\mathbf{Inst}\rightarrow\mathbf{Set}.$
fong-spivak-seven-sketches_1 EQ0020 (2.10)	55	■ 0.266	$\begin{array}{l} h_{\mathfrak{1}}:=\{2\diamond,3\diamond,10\bullet,J\bullet,Q\bullet\}\quad i_{\mathfrak{1}}:=\{4\diamond,4\diamond,6\diamond,6\diamond,10\diamond\}\\ \bullet,J\bullet,Q\bullet\}\quad i_{\mathfrak{2}}:=\{5\diamond,5\diamond,7\diamond,J\diamond,Q\diamond\}.\\ h_{\mathfrak{2}}:=\{2\diamond,3\diamond,4\diamond,K\star,A\diamond\}\quad i_{\mathfrak{2}}:=\{5\diamond,5\diamond,7\diamond,J\diamond,Q\diamond\}. \end{array}$	$h_1:=\{2\diamond,3\diamond,10\bullet,J\bullet,Q\bullet\}\quad i_1:=\{4\diamond,4\diamond,6\diamond,6\diamond,10\diamond\}$ $h_2:=\{2\diamond,3\diamond,4\diamond,K\star,A\diamond\}\quad i_2:=\{5\diamond,5\diamond,7\diamond,J\diamond,Q\diamond\}.$	$h_1:=\{2\heartsuit,3\heartsuit,10\spadesuit,J\spadesuit,Q\spadesuit\}\quad i_1:=\{4\clubsuit,4\clubsuit,6\heartsuit,6\heartsuit,10\diamond\}$ $h_2:=\{2\diamond,3\diamond,4\diamond,K\spadesuit,A\spadesuit\}\quad i_2:=\{5\spadesuit,5\heartsuit,7\heartsuit,J\diamond,Q\diamond\}.$
fong-spivak-seven-sketches_1 EQ0033	73	■ 0.268	$X:=\operatorname{Ob}(\mathfrak{X})\quad d(x,y):=\mathfrak{X}(x,y).$	$X:=\operatorname{Ob}(\mathfrak{X})\quad d(x,y):=\mathfrak{X}(x,y).$	$X:=\operatorname{Ob}(\mathfrak{X})\quad d(x,y):=\mathfrak{X}(x,y).$

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fong-spivak-seven-sketches_1 EQ0068	115	■ 0.352	$\mathcal{C} \underset{R}{\stackrel{L}{\Longleftarrow\longrightarrow}} \mathcal{D}$	$\mathcal{C} \overset{L}{\underset{R}{\rightleftarrows}} \mathcal{D}$	
fong-spivak-seven-sketches_1 EQ0206	290	■ 0.362	$A, \quad A ; f, \quad A ; g, \quad A ; j, \quad B, \quad B ; h, \quad C, \quad C ; i, \quad D$	$A, \quad A;f, \quad A;g, \quad A;j, \quad B, \quad B;h, \quad C, \quad C;i, \quad D$	$A, \quad A \circ f, \quad A \circ g, \quad A \circ j, \quad B, \quad B \circ h, \quad C, \quad C \circ i, \quad D$
fong-spivak-seven-sketches_1 EQ0098	156	■ 0.366	$(x \times y) \left((x, y), \left(x^{\prime}, y^{\prime} \right) \right) := x \left(x, x^{\prime} \right) \otimes y \left(y, y^{\prime} \right) .$	$(x \times y) \left((x, y), (x', y') \right) := x(x, x') \otimes y(y, y') .$	$(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y') .$
fong-spivak-seven-sketches_1 EQ0036	77	■ 0.376	$\begin{array}{l} \mathcal{C}_{\mathcal{f}}(c, d) \otimes_W \mathcal{C}_{\mathcal{f}}(d, e) \leq f(\mathcal{C}(c, d) \otimes_V \mathcal{C}(d, e)) \\ f(\mathcal{C}(c, d) \otimes_V \mathcal{C}(d, e)) \leq f(\mathcal{C}(c, e)) \\ \mathcal{C}_{\mathcal{f}}(c, e) \end{array}$	$\begin{array}{l} \mathcal{C}_{\mathcal{f}}(c, d) \otimes_W \mathcal{C}_{\mathcal{f}}(d, e) = f(\mathcal{C}(c, d) \otimes_W f(\mathcal{C}(d, e))) \\ \leq f(\mathcal{C}(c, d) \otimes_V \mathcal{C}(d, e)) \\ \leq f(\mathcal{C}(c, e)) \\ = \mathcal{C}_{\mathcal{f}}(c, e) \end{array}$	$\begin{array}{l} \mathcal{C}_{\mathcal{f}}(c, d) \otimes_W \mathcal{C}_{\mathcal{f}}(d, e) = f(\mathcal{C}(c, d)) \otimes_W f(\mathcal{C}(d, e)) \\ \leq f(\mathcal{C}(c, d) \otimes_V \mathcal{C}(d, e)) \\ \leq f(\mathcal{C}(c, e)) \\ = \mathcal{C}_{\mathcal{f}}(c, e) \end{array}$
fong-spivak-seven-sketches_1 EQ0142	227	■ 0.424	$(g \circ f) \left(x_1, \dots, x_{i-1}, w_1, \dots, w_m, x_{i+1}, \dots, x_n \right) = g \left(x_1, \dots, x_{i-1}, f \left(w_1, \dots, w_m \right), x_{i+1}, \dots, x_n \right)$	$(g \circ f) \left(x_1, \dots, x_{i-1}, w_1, \dots, w_m, x_{i+1}, \dots, x_n \right) = g \left(x_1, \dots, x_{i-1}, f \left(w_1, \dots, w_m \right), x_{i+1}, \dots, x_n \right)$	$(g \circ_i f)(x_1, \dots, x_{i-1}, w_1, \dots, w_m, x_{i+1}, \dots, x_n) = g(x_1, \dots, x_{i-1}, f(w_1, \dots, w_m), x_{i+1}, \dots, x_n)$
fong-spivak-seven-sketches_1 EQ0228	303	■ 0.461	$(\alpha^{-1}; \alpha)(x, z) = \bigvee_{(y, 1) \in \mathcal{X} \times \mathbf{1}} \alpha(x, (y, 1)) \otimes \alpha^{-1}((y, 1), z) = \bigvee_{y \in \mathcal{X}} x(x, y) \otimes x(y, z) = x(x, z).$	$(\alpha^{-1}; \alpha)(x, z) = \bigvee_{(y, 1) \in \mathcal{X} \times \mathbf{1}} \alpha(x, (y, 1)) \otimes \alpha^{-1}((y, 1), z) = \bigvee_{y \in \mathcal{X}} x(x, y) \otimes x(y, z) = x(x, z).$	$(\alpha^{-1} \circ \alpha)(x, z) = \bigvee_{(y, 1) \in \mathcal{X} \times \mathbf{1}} \alpha(x, (y, 1)) \otimes \alpha^{-1}((y, 1), z) = \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) = \mathcal{X}(x, z).$
fong-spivak-seven-sketches_1 EQ0227	303	■ 0.463	$\bigvee_{y \in \mathcal{X}} x(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} x(x, z) = \mathcal{X}(x, z).$	$\bigvee_{y \in \mathcal{X}} x(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} x(x, z) = \mathcal{X}(x, z).$	$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$
fong-spivak-seven-sketches_1 EQ0071 (3.75)	118	■ 0.470	$! : \mathcal{C} \rightarrow \mathbf{1} .$	$! : \mathcal{C} \rightarrow \mathbf{1} .$	$! : \mathcal{C} \rightarrow \mathbf{1} .$
fong-spivak-seven-sketches_1 EQ0039 (2.77)	79	■ 0.472	$(\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_2, y_2) \right) \otimes (\mathcal{X} \times \mathcal{Y}) \left((x_2, y_2), (x_3, y_3) \right) \leq (\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_3, y_3) \right) .$	$(\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_2, y_2) \right) \otimes (\mathcal{X} \times \mathcal{Y}) \left((x_2, y_2), (x_3, y_3) \right) \leq (\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_3, y_3) \right) .$	$(\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_2, y_2) \right) \otimes (\mathcal{X} \times \mathcal{Y}) \left((x_2, y_2), (x_3, y_3) \right) \leq (\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_3, y_3) \right) .$
fong-spivak-seven-sketches_1 EQ0163	257	■ 0.487	$\Omega_{\text{Grph}} = (0, \emptyset ; \emptyset) \subset \emptyset \underset{\{(V, V ; A)\}^{\{(V, V ; \emptyset)\}}}{\stackrel{\{(0, V ; \emptyset)\}}{\longrightarrow}} \{ \underset{\{(V, V ; A)\}^{\{(V, V ; \emptyset)\}}}{\stackrel{\{(0, V ; \emptyset)\}}{\longrightarrow}} \}$	$\Omega_{\text{Grph}} = (0, 0; 0) \subset 0 \overset{(0, V; 0)}{\underset{(V, 0; 0)}{\rightleftarrows}} V \overset{(V, V; 0)}{\underset{(V, V; A)}{\rightleftarrows}} V$	

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fong-spivak-seven-sketches-EQ0258	317	■ 0.494	$\left[\mathrm{id}_{\{A\}},!_{\{A\}}\right]\stackrel{\circ}{\circlearrowleft}\{9\}$ $\mathrm{iota}_{\{A\}}=\left[\mathrm{id}_{\{A\}}\stackrel{\circ}{\circlearrowleft}\{9\}\right]$ $\mathrm{iota}_{\{A\}},!_{\{A\}}^{\circlearrowleft}\stackrel{\circlearrowleft}{\circlearrowleft}_{\{A\}}$ $\mathrm{iota}_{\{A\}}\right]=\left[\mathrm{iota}_{\{A\}},!_{\{A+\varnothing\}}\right]$ $\right]=\left[\mathrm{iota}_{\{A\}},\mathrm{iota}_{\{\varnothing\}}\right]=$ $\mathrm{id}_{\{A+\varnothing\}}.$	$[\mathrm{id}_A,!_A]\overset{\circ}{\circlearrowleft}\iota_A = \left[\mathrm{id}_A\overset{\circ}{\circlearrowleft}\iota_A,!_A^{\circlearrowleft}\overset{\circ}{\circlearrowleft}\iota_A\right] = [\iota_A,!_{A+\varnothing}] = [\iota_A,\iota_{\varnothing}] = \mathrm{id}_{A+\varnothing}.$	$[\mathrm{id}_A,!_A]\overset{\circ}{\circlearrowleft}\iota_A = [\mathrm{id}_A\overset{\circ}{\circlearrowleft}\iota_A,!_A^{\circlearrowleft}\overset{\circ}{\circlearrowleft}\iota_A] = [\iota_A,!_{A+\varnothing}] = [\iota_A,\iota_{\varnothing}] = \mathrm{id}_{A+\varnothing}.$
fong-spivak-seven-sketches-EQ0074	119	■ 0.500	$\begin{array}{ c } \hline \text{ Bob-Grace-Pat-Emmy } \\ \hline \end{array}$ $\text{ Sue-Doug } \end{array}$	$\frac{1}{\begin{array}{ c } \hline \text{ Bob-Grace-Pat-Emmy } \\ \hline \end{array} \text{ Sue-Doug }}$	$\frac{1}{\begin{array}{ c } \hline \text{ Bob-Grace-Pat-Emmy } \\ \hline \end{array} \text{ Sue-Doug }}$
fong-spivak-seven-sketches-EQ0223	299	■ 0.512	$\begin{aligned} \Phi &= M_X^3 * M_{\Phi} * M_Y^2 = \\ &= \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \end{aligned}$	$\Phi = M_X^3 * M_{\Phi} * M_Y^2 = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix}$ $= \begin{pmatrix} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix}$ $= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix}$	$\Phi = M_X^3 * M_{\Phi} * M_Y^2 = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix}$ $= \begin{pmatrix} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{pmatrix} \begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix}$ $= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix}$
fong-spivak-seven-sketches-EQ0021	58	■ 0.515	$\left.\left.\left.\frac{\bar{x}}{y}\right.\right.\right)=\frac{x\otimes y}{z}=\left\{\frac{x}{y\otimes z}\right\}=\left\{\frac{y}{z}\right\}$	$\left.\left.\left.\frac{\bar{x}}{y}\right.\right.\right)=\frac{x\otimes y}{z}=\left\{\frac{x}{y\otimes z}\right\}=\left\{\frac{y}{z}\right\}$	$\left.\left.\frac{x}{y}\right.\right)=\frac{x\otimes y}{z}=\left\{\frac{x}{y\otimes z}=\left\{\frac{y}{z}\right\}\right\}$
fong-spivak-seven-sketches-EQ0069	115	■ 0.520	$\text{Set}\underset{(-)^B}{\overset{-\times B}{\Rrightarrow}}\text{Set}\quad\text{Set}(A\times B,C)\cong\text{Set}(A,C^B)$	$\text{Set}\overset{-\times B}{\underset{(-)^B}{\Rrightarrow}}\text{Set}\quad\text{Set}(A\times B,C)\cong\text{Set}(A,C^B)$	$\text{Set}\overset{-\times B}{\underset{(-)^B}{\Rrightarrow}}\text{Set}\quad\text{Set}(A\times B,C)\cong\text{Set}(A,C^B)$
fong-spivak-seven-sketches-EQ0116	186	■ 0.520	$\mathrm{B}(\neg\mathrm{C})=\{(x,(x,x))\mid x\in R\}.$	$\mathrm{B}(\neg C) = \{(x,(x,x)) \mid x \in R\}.$	$\mathrm{B}(\neg C) = \{(x,(x,x)) \mid x \in R\}.$
fong-spivak-seven-sketches-EQ0257	317	■ 0.522	$\mathrm{id}_{\{A\}}+\mathrm{id}_{\{B\}}=\left[\mathrm{id}_{\{A\}}\stackrel{\circ}{\circlearrowleft}\mathrm{iota}_{\{A\}},\mathrm{id}_{\{B\}}\stackrel{\circ}{\circlearrowleft}\mathrm{iota}_{\{B\}}\right]=\left[\mathrm{iota}_{\{A\}},\mathrm{iota}_{\{B\}}\right]=\mathrm{id}_{\{A+B\}}.$	$\mathrm{id}_A+\mathrm{id}_B = \left[\mathrm{id}_A\overset{\circ}{\circlearrowleft}\iota_A,\mathrm{id}_B\overset{\circ}{\circlearrowleft}\iota_B\right] = [\iota_A,\iota_B] = \mathrm{id}_{A+B}.$	$\mathrm{id}_A+\mathrm{id}_B = [\mathrm{id}_A\overset{\circ}{\circlearrowleft}\iota_A,\mathrm{id}_B\overset{\circ}{\circlearrowleft}\iota_B] = [\iota_A,\iota_B] = \mathrm{id}_{A+B}.$

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fong-spivak-seven-sketches EQ0054 (3.11)	95	0.526	$\mathbf{3} := \operatorname{Free} \left(\left(\begin{array}{c} v_1 \\ \bullet \end{array} \right) \xrightarrow{f_1} \left(\begin{array}{c} v_2 \\ \bullet \end{array} \right) \xrightarrow{f_2} \left(\begin{array}{c} v_3 \\ \bullet \end{array} \right) \right)$	$\mathbf{3} := \operatorname{Free} \left(\begin{array}{c} v_1 \\ \bullet \end{array} \xrightarrow{f_1} \begin{array}{c} v_2 \\ \bullet \end{array} \xrightarrow{f_2} \begin{array}{c} v_3 \\ \bullet \end{array} \right)$	$\mathbf{3} := \operatorname{Free} \left(\begin{array}{c} v_1 \\ \bullet \end{array} \xrightarrow{f_1} \begin{array}{c} v_2 \\ \bullet \end{array} \xrightarrow{f_2} \begin{array}{c} v_3 \\ \bullet \end{array} \right)$
fong-spivak-seven-sketches EQ0005	33	0.539	$p = g(f(p)) \quad \text{and} \quad q = f(g(q))$	$p = g(f(p)) \quad \text{and} \quad q = f(g(q))$	$p = g(f(p)) \quad \text{and} \quad q = f(g(q))$
fong-spivak-seven-sketches EQ0078 (4.3)	132	0.549	$\Phi: x^{\operatorname{op}} \times y \rightarrow \operatorname{Bool}$	$\Phi: x^{\operatorname{op}} \times y \rightarrow \operatorname{Bool}$	$\Phi: \mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \rightarrow \mathbf{Bool}$
fong-spivak-seven-sketches EQ0091	142	0.550	$\widehat{F}(p, q) := Q(F(p), q) \quad \text{and} \quad \check{F}(q, p) := Q(q, F(p))$	$\widehat{F}(p, q) := Q(F(p), q) \quad \text{and} \quad \check{F}(q, p) := Q(q, F(p))$	$\widehat{F}(p, q) := Q(F(p), q) \quad \text{and} \quad \check{F}(q, p) := Q(q, F(p))$
fong-spivak-seven-sketches EQ0104 (5.17)	165	0.566	$G + G' := ((V \sqcup V'), [\text{in}, \text{in}'], [\text{out}, \text{out}'], (\iota \sqcup \iota'))$	$G + G' := ((V \sqcup V'), [\text{in}, \text{in}'], [\text{out}, \text{out}'], (\iota \sqcup \iota'))$	$G + G' := ((V \sqcup V'), [\text{in}, \text{in}'], [\text{out}, \text{out}'], (\iota \sqcup \iota'))$
fong-spivak-seven-sketches EQ0221	298	0.568	$x \left(x^{\prime}, x \right) \otimes y \left(y, y^{\prime} \right) \leq \Phi(x, y) \multimap \Phi(x', y')$	$x(x', x) \otimes y(y, y') \leq \Phi(x, y) \multimap \Phi(x', y')$	$\mathcal{X}(x', x) \otimes \mathcal{Y}(y, y') \leq \Phi(x, y) \multimap \Phi(x', y')$
fong-spivak-seven-sketches EQ0253	313	0.570	$\left\{ (y, z) \mid \text{there exists } x \in R^m \text{ such that } y = S(g)(x) \text{ and } S(h)(x) = z \right\}$	$\{(y, z) \mid \text{there exists } x \in R^m \text{ such that } y = S(g)(x) \text{ and } S(h)(x) = z\}$	$\{(y, z) \mid \text{there exists } x \in R^m \text{ such that } y = S(g)(x) \text{ and } S(h)(x) = z\}$
fong-spivak-seven-sketches EQ0043	82	0.593	$2 \operatorname{H}_2 \rightarrow \operatorname{H}_2 \operatorname{O} \rightarrow (2 \operatorname{Na} \rightarrow (2 \operatorname{NaOH} + \operatorname{H}_2))$	$2\operatorname{H}_2\operatorname{O} \rightarrow (2\operatorname{Na} \rightarrow (2\operatorname{NaOH} + \operatorname{H}_2))$	$2\operatorname{H}_2\operatorname{O} \rightarrow (2\operatorname{Na} \multimap (2\operatorname{NaOH} + \operatorname{H}_2))$
fong-spivak-seven-sketches EQ0179	272	0.598	$h, \quad 1, \quad 2, \quad 3$	$h, \quad 1, \quad 2, \quad 3$	$h, \quad 1, \quad 2, \quad 3$
fong-spivak-seven-sketches EQ0002 (1.15)	21	0.658	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset$	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset$	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset$
fong-spivak-seven-sketches EQ0158 (7.37)	250	0.658	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s; f)(u) = u \text{ for all } u \in U\}$	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s; f)(u) = u \text{ for all } u \in U\}$	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s \circ f)(u) = u \text{ for all } u \in U\}$
fong-spivak-seven-sketches EQ0147 (7.2)	235	0.661	$((A = \operatorname{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \operatorname{stop}) \Rightarrow \ddot{S} = 0)$	$((A = \operatorname{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \operatorname{stop}) \Rightarrow \ddot{S} = 0)$	$((A = \operatorname{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \operatorname{stop}) \Rightarrow \ddot{S} = 0)$
fong-spivak-seven-sketches EQ0191	282	0.677	$\begin{array}{l} t + u \leq (v + w) + u \quad (\text{monotonicity, } t \leq v + w, u \leq u) \\ = v + (w + u) \quad (\text{associativity}) \\ \leq v + (x + z) \quad (\text{monotonicity, } v \leq v, w + u \leq x + v) \\ = (v + x) + z \quad (\text{associativity}) \\ \leq y + z. \quad (\text{monotonicity, } v + x \leq y, z \leq z) \end{array}$	$\begin{array}{l} t + u \leq (v + w) + u \quad (\text{monotonicity, } t \leq v + w, u \leq u) \\ = v + (w + u) \quad (\text{associativity}) \\ \leq v + (x + z) \quad (\text{monotonicity, } v \leq v, w + u \leq x + v) \\ = (v + x) + z \quad (\text{associativity}) \\ \leq y + z. \quad (\text{monotonicity, } v + x \leq y, z \leq z) \end{array}$	$\begin{array}{l} t + u \leq (v + w) + u \quad (\text{monotonicity, } t \leq v + w, u \leq u) \\ = v + (w + u) \quad (\text{associativity}) \\ \leq v + (x + z) \quad (\text{monotonicity, } v \leq v, w + u \leq x + v) \\ = (v + x) + z \quad (\text{associativity}) \\ \leq y + z. \quad (\text{monotonicity, } v + x \leq y, z \leq z) \end{array}$

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fong-spivak-seven-sketches EQ0224	300	0.684	$\begin{aligned} \Phi: \Psi &= \left(M_X^3 * M_\Phi * M_Y^2 \right) * \left(M_Y^2 * M_\Psi * M_Z^3 \right) \\ &= M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ &= \left(M_X^3 * M_\Phi * M_Y^2 \right) * M_\Psi * M_Z^3 \\ &= \Phi * M_\Psi * M_Z^3 \end{aligned}$	$\Phi: \Psi = \left(M_X^3 * M_\Phi * M_Y^2 \right) * \left(M_Y^2 * M_\Psi * M_Z^3 \right) \\ = M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ = \left(M_X^3 * M_\Phi * M_Y^2 \right) * M_\Psi * M_Z^3 \\ = \Phi * M_\Psi * M_Z^3$	$\Phi \circ \Psi = (M_X^3 * M_\Phi * M_Y^2) * (M_Y^2 * M_\Psi * M_Z^3) \\ = M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ = (M_X^3 * M_\Phi * M_Y^2) * M_\Psi * M_Z^3 \\ = \Phi * M_\Psi * M_Z^3$
fong-spivak-seven-sketches EQ0120 (5.81)	187	0.690	$\mathbf{SFG}_R^+ := \text{Free} \left(G_R \sqcup G_R^{\text{op}} \right).$	$\mathbf{SFG}_R^+ := \text{Free} \left(G_R \sqcup G_R^{\text{op}} \right).$	$\mathbf{SFG}_R^+ := \text{Free} \left(G_R \sqcup G_R^{\text{op}} \right).$
fong-spivak-seven-sketches EQ0081 (◁)	133	0.690	$x \left(x', x \right) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$x \left(x', x \right) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes \mathcal{Y}(y, y') \leq \Phi(x', y').$
fong-spivak-seven-sketches EQ0140 (6.81)	220	0.725	$\begin{aligned} \psi_{V,V'}: \text{Circ}(V) \times \text{Circ}(V') &\longrightarrow \text{Circ}(V + V'); \\ ((V, A, s, t, \ell), (V', A', s', t', \ell')) &\longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']). \end{aligned}$	$\psi_{V,V'}: \text{Circ}(V) \times \text{Circ}(V') \longrightarrow \text{Circ}(V + V'); \\ ((V, A, s, t, \ell), (V', A', s', t', \ell')) \longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']).$	$\psi_{V,V'}: \text{Circ}(V) \times \text{Circ}(V') \longrightarrow \text{Circ}(V + V'); \\ ((V, A, s, t, \ell), (V', A', s', t', \ell')) \longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']).$
fong-spivak-seven-sketches EQ0222	298	0.731	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes \mathcal{Y}(y, y') \leq \Phi(x', y').$

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fong-spivak-seven-sketches-EQ0132	212	0.742	$\begin{array}{rrrr} \operatorname{in}(\mu) := 2, & \operatorname{in}(\eta) := 0, & \operatorname{in}(\delta) := 1, & \operatorname{in}(\epsilon) := 1, \\ \operatorname{out}(\mu) := 1, & \operatorname{out}(\eta) := 1, & \operatorname{out}(\delta) := 2, & \operatorname{out}(\epsilon) := 0. \end{array}$	$\begin{array}{rrrr} \operatorname{in}(\mu) := 2, & \operatorname{in}(\eta) := 0, & \operatorname{in}(\delta) := 1, & \operatorname{in}(\epsilon) := 1, \\ \operatorname{out}(\mu) := 1, & \operatorname{out}(\eta) := 1, & \operatorname{out}(\delta) := 2, & \operatorname{out}(\epsilon) := 0. \end{array}$	$\begin{array}{rrrr} \operatorname{in}(\mu) := 2, & \operatorname{in}(\eta) := 0, & \operatorname{in}(\delta) := 1, & \operatorname{in}(\epsilon) := 1, \\ \operatorname{out}(\mu) := 1, & \operatorname{out}(\eta) := 1, & \operatorname{out}(\delta) := 2, & \operatorname{out}(\epsilon) := 0. \end{array}$
fong-spivak-seven-sketches-EQ0097	155	0.755	$\backslash[\alpha: \beta := \iota_{A \sqcup C}^* ((\iota_{A \sqcup B})!(\alpha) \vee (\iota_{B \sqcup C})!(\beta)).$	$\alpha : \beta := \iota_{A \sqcup C}^* ((\iota_{A \sqcup B})!(\alpha) \vee (\iota_{B \sqcup C})!(\beta)).$	$\alpha \circ \beta := \iota_{A \sqcup C}^* ((\iota_{A \sqcup B})!(\alpha) \vee (\iota_{B \sqcup C})!(\beta)).$
fong-spivak-seven-sketches-EQ0230	304	0.759	$\begin{aligned} & \bigvee_{a,b,c,d,e \in X} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_X(a,b) \otimes \eta_X(1,c,d) \otimes \epsilon_X(b,c,1) \otimes \mathrm{U}_X(d,e) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in X} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_X(a,b) \otimes \eta_X(1,c,d) \otimes \epsilon_X(b,c,1) \otimes \mathrm{U}_X(d,e) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in X} X(x,a) \otimes X(a,b) \otimes X(c,d) \otimes X(b,c) \otimes X(d,e) \otimes X(e,y) \\ & = X(x,y) \end{aligned}$	$\begin{aligned} & \bigvee_{a,b,c,d,e \in X} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_X(a,b) \otimes \eta_X(1,c,d) \otimes \epsilon_X(b,c,1) \otimes \mathrm{U}_X(d,e) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in X} X(x,a) \otimes X(a,b) \otimes X(c,d) \otimes X(b,c) \otimes X(d,e) \otimes X(e,y) \\ & = X(x,y) \end{aligned}$	$\begin{aligned} & \bigvee_{a,b,c,d,e \in X} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_X(a,b) \otimes \eta_X(1,c,d) \otimes \epsilon_X(b,c,1) \otimes \mathrm{U}_X(d,e) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in X} X(x,a) \otimes X(a,b) \otimes X(c,d) \otimes X(b,c) \otimes X(d,e) \otimes X(e,y) \\ & = X(x,y) \end{aligned}$
fong-spivak-seven-sketches-EQ0242	310	0.775	$c_n := -c : (1 + -c) : (1 + 1 + -c) : \cdots : (1 + \cdots + 1 + -c) : 1 \rightarrow n$	$c_n := -c : (1 + -c) : (1 + 1 + -c) : \cdots : (1 + \cdots + 1 + -c) : 1 \rightarrow n$	$c_n := -c : (1 + -c) : (1 + 1 + -c) : \cdots : (1 + \cdots + 1 + -c) : 1 \rightarrow n$
fong-spivak-seven-sketches-EQ0156	248	0.780	$\mathbf{Op}_1 := \{\emptyset, \{1\}, X\} \quad \text{and} \quad \mathbf{Op}_2 := \{\emptyset, \{2\}, X\}$	$\mathbf{Op}_1 := \{\emptyset, \{1\}, X\} \quad \text{and} \quad \mathbf{Op}_2 := \{\emptyset, \{2\}, X\}$	$\mathbf{Op}_1 := \{\emptyset, \{1\}, X\} \quad \text{and} \quad \mathbf{Op}_2 := \{\emptyset, \{2\}, X\}$
fong-spivak-seven-sketches-EQ0219	298	0.795	$\begin{aligned} & \Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book}) \\ & = \Lambda(\text{preorder}, \text{this book}) = \Lambda(\text{category}, \text{this book}) = \text{true} \\ & \Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{category}, \text{nothing}) = \text{false}. \end{aligned}$	$\begin{aligned} & \Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book}) \\ & = \Lambda(\text{preorder}, \text{this book}) = \Lambda(\text{category}, \text{this book}) = \text{true} \\ & \Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{category}, \text{nothing}) = \text{false}. \end{aligned}$	$\begin{aligned} & \Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book}) \\ & = \Lambda(\text{preorder}, \text{this book}) = \Lambda(\text{category}, \text{this book}) = \text{true} \\ & \Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{category}, \text{nothing}) = \text{false}. \end{aligned}$
fong-spivak-seven-sketches-EQ0077	124	0.797	$\begin{aligned} & \lim_{\mathcal{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ & \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}. \end{aligned}$	$\lim_{\mathcal{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}.$	$\lim_{\mathcal{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}.$
fong-spivak-seven-sketches-EQ0131	211	0.808	$m \text{ legs } \lesssim n \text{ legs}$	$m \text{ legs } \lesssim n \text{ legs}$	$m \text{ legs } \left\{ \begin{array}{c} \text{diagram} \end{array} \right\} n \text{ legs}$

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fong-spivak-seven-sketches-1 EQ0193	286	0.825	$\begin{aligned} (x \times y) \left(\left(x_1, y_1 \right), \left(x_2, y_2 \right) \right) \otimes (x \times y) \left(\left(x_2, y_2 \right), \left(x_3, y_3 \right) \right) \\ = x \left(x_1, x_2 \right) \otimes y \left(y_1, y_2 \right) \otimes x \left(x_2, x_3 \right) \otimes y \left(y_2, y_3 \right) \\ = x \left(x_1, x_2 \right) \otimes x \left(x_2, x_3 \right) \otimes y \left(y_1, y_2 \right) \otimes y \left(y_2, y_3 \right) \\ \leq x \left(x_1, x_3 \right) \otimes y \left(y_1, y_3 \right) \\ = (x \times y) \left(\left(x_1, y_1 \right), \left(x_3, y_3 \right) \right). \end{aligned}$	$(x \times y) \left(\left(x_1, y_1 \right), \left(x_2, y_2 \right) \right) \otimes (x \times y) \left(\left(x_2, y_2 \right), \left(x_3, y_3 \right) \right)$ $= x \left(x_1, x_2 \right) \otimes y \left(y_1, y_2 \right) \otimes x \left(x_2, x_3 \right) \otimes y \left(y_2, y_3 \right)$ $= x \left(x_1, x_2 \right) \otimes x \left(x_2, x_3 \right) \otimes y \left(y_1, y_2 \right) \otimes y \left(y_2, y_3 \right)$ $\leq x \left(x_1, x_3 \right) \otimes y \left(y_1, y_3 \right)$ $= (x \times y) \left(\left(x_1, y_1 \right), \left(x_3, y_3 \right) \right).$	$(\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_2, y_2)) \otimes (\mathcal{X} \times \mathcal{Y})((x_2, y_2), (x_3, y_3))$ $= \mathcal{X}(x_1, x_2) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_2, y_3)$ $= \mathcal{X}(x_1, x_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{Y}(y_2, y_3)$ $\leq \mathcal{X}(x_1, x_3) \otimes \mathcal{Y}(y_1, y_3)$ $= (\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_3, y_3)).$
fong-spivak-seven-sketches-1 EQ0172	264	0.827	$\mathbb{I} \mid \mathbb{R} := \{ [d, u] \mid d \leq u \}.$	$\mathbb{I} \mathbb{R} := \{ [d, u] \subseteq \mathbb{R} \mid d \leq u \}.$	$\mathbb{I} \mathbb{R} := \{ [d, u] \subseteq \mathbb{R} \mid d \leq u \}.$
fong-spivak-seven-sketches-1 EQ0093	143	0.828	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} x(a, b) & \text{if } a, b \in X \\ \Phi(a, b) & \text{if } a \in X, b \in Y \\ \emptyset & \text{if } a \in Y, b \in X \\ y(a, b) & \text{if } a, b \in Y \end{cases}$	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} x(a, b) & \text{if } a, b \in X \\ \Phi(a, b) & \text{if } a \in X, b \in Y \\ \emptyset & \text{if } a \in Y, b \in X \\ y(a, b) & \text{if } a, b \in Y \end{cases}$	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} \mathcal{X}(a, b) & \text{if } a, b \in \mathcal{X} \\ \Phi(a, b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \emptyset & \text{if } a \in \mathcal{Y}, b \in \mathcal{X} \\ \mathcal{Y}(a, b) & \text{if } a, b \in \mathcal{Y} \end{cases}$
fong-spivak-seven-sketches-1 EQ0022	58	0.830	$\frac{\frac{x}{y}}{z} = \frac{\frac{x}{y}}{\frac{z}{z}}$	$\frac{\frac{x}{y}}{z} = \frac{\frac{x}{y}}{\frac{z}{z}}$	$\frac{\frac{x}{y}}{z} = \frac{\frac{x}{y}}{\frac{z}{z}}$
fong-spivak-seven-sketches-1 EQ0146	229	0.839	$\operatorname{Circ}(\varphi) : \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$	$\operatorname{Circ}(\varphi) : \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$	$\operatorname{Circ}(\varphi) : \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$
fong-spivak-seven-sketches-1 EQ0086 (4.25)	140	0.842	$\operatorname{U}_x(x, y) := \mathcal{X}(x, y).$	$\operatorname{U}_x(x, y) := \mathcal{X}(x, y).$	$\operatorname{U}_x(x, y) := \mathcal{X}(x, y).$
fong-spivak-seven-sketches-1 EQ0136	219	0.843	$C := \{ \text{light}, \text{switch}, \text{battery} \} \sqcup \{ x\Omega \mid x \in \mathbb{R}^+ \}.$	$C := \{ \text{light}, \text{switch}, \text{battery} \} \sqcup \{ x\Omega \mid x \in \mathbb{R}^+ \}.$	$C := \{ \text{light}, \text{switch}, \text{battery} \} \sqcup \{ x\Omega \mid x \in \mathbb{R}^+ \}.$
fong-spivak-seven-sketches-1 EQ0125	197	0.843	$\mathcal{C} := \stackrel{A}{\bullet} \stackrel{f}{\rightrightarrows} \stackrel{B}{\bullet}$	$\mathcal{C} := \stackrel{A}{\bullet} \stackrel{f}{\rightrightarrows} \stackrel{B}{\bullet}$	$\mathcal{C} := \stackrel{A}{\bullet} \stackrel{f}{\rightrightarrows} \stackrel{B}{\bullet}$
fong-spivak-seven-sketches-1 EQ0003	24	0.849	$S := \begin{array}{ ccc } \hline 11 & 12 & 13 \\ \bullet & & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \\ \hline \end{array}$	$S := \begin{array}{ ccc } \hline 11 & 12 & 13 \\ \bullet & & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \\ \hline \end{array}$	$S := \begin{array}{ ccc } \hline 11 & 12 & 13 \\ \bullet & & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \\ \hline \end{array}$
fong-spivak-seven-sketches-1 EQ0121	188	0.868	$\operatorname{B}(g \circ (h^{\operatorname{op}})) = \{ (x, y) \mid S(g)(x) = S(h)(y) \}.$	$\operatorname{B}(g \circ (h^{\operatorname{op}})) = \{ (x, y) \mid S(g)(x) = S(h)(y) \}.$	$\operatorname{B}(g \circ (h^{\operatorname{op}})) = \{ (x, y) \mid S(g)(x) = S(h)(y) \}.$

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fong-spivak-seven-sketches-EQ0107	174	0.873	$\begin{array}{r} \dot{x}+3\ddot{y}-2z=0 \\ \dot{z}=0 \end{array}$	$\begin{array}{l} \dot{x}+3\ddot{y}-2z=0 \\ \ddot{y}+5\dot{z}=0 \end{array}$	$\begin{array}{l} \dot{x}+3\ddot{y}-2z=0 \\ \ddot{y}+5\dot{z}=0 \end{array}$
fong-spivak-seven-sketches-EQ0010 (1.108)	42	0.877	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$
fong-spivak-seven-sketches-EQ0079 (4.5)	132	0.881	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$
fong-spivak-seven-sketches-EQ0217	296	0.890	$\begin{aligned} \lim_{\mathfrak{J}} \{D\} &:= \left\{ \left(d_1, \dots, d_n \right) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \right. \\ &\quad \left. \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j \right\}. \end{aligned}$	$\lim_{\mathfrak{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}.$	$\lim_{\mathfrak{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}.$
fong-spivak-seven-sketches-EQ0153 (7.18)	242	0.894	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true}), (\text{false}, \text{false})\} \subseteq \mathbb{B} \times \mathbb{B}$	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$
fong-spivak-seven-sketches-EQ0105	167	0.897	$a \subset \bullet \supset b$	$a \subset \bullet \supset b$	$a \curvearrowright \bullet \curvearrowleft b$
fong-spivak-seven-sketches-EQ0143 (6.95)	227	0.902	$\left(g \circ \text{circ}\right)_i: X_1 \times \cdots \times X_{i-1} \times W_1 \times \cdots \times W_m \times X_{i+1} \times \cdots \times X_n \longrightarrow Y$	$(g \circ_i f): X_1 \times \cdots \times X_{i-1} \times W_1 \times \cdots \times W_m \times X_{i+1} \times \cdots \times X_n \longrightarrow Y$	$(g \circ_i f): X_1 \times \cdots \times X_{i-1} \times W_1 \times \cdots \times W_m \times X_{i+1} \times \cdots \times X_n \longrightarrow Y$
fong-spivak-seven-sketches-EQ0129	205	0.913	$\left\{ (v, d) \mid d \in D(v), \text{ where } v = v_1 \text{ or } v = v_2 \right\}.$	$\{(v, d) \mid d \in D(v), \text{ where } v = v_1 \text{ or } v = v_2\}.$	$\{(v, d) \mid d \in D(v), \text{ where } v = v_1 \text{ or } v = v_2\}.$
fong-spivak-seven-sketches-EQ0164	257	0.917	$\stackrel{A}{\longrightarrow} \bullet \stackrel{B}{\longrightarrow} \bullet \stackrel{C}{\longrightarrow} \bullet$	$\stackrel{A}{\longrightarrow} \bullet \stackrel{B}{\longrightarrow} \bullet \stackrel{C}{\longrightarrow} \bullet$	$\stackrel{A}{\longrightarrow} \bullet \stackrel{B}{\longrightarrow} \bullet \stackrel{C}{\longrightarrow} \bullet$
fong-spivak-seven-sketches-EQ0256	315	0.919	$\begin{aligned} f(m) *_{\mathbb{R}} f(n) &= \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{m \text{ summands}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} + \cdots + 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= \underbrace{1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}}}_{mn \text{ summands}} = f(m * n). \end{aligned}$	$\begin{aligned} f(m) *_{\mathbb{R}} f(n) &= \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{m \text{ summands}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} + \cdots + 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= \underbrace{1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}}}_{mn \text{ summands}} = f(m * n). \end{aligned}$	$\begin{aligned} f(m) *_{\mathbb{R}} f(n) &= \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{m \text{ summands}} *_{\mathbb{R}} \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} + \cdots + 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= \underbrace{1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}}}_{mn \text{ summands}} = f(m * n). \end{aligned}$
fong-spivak-seven-sketches-EQ0096	149	0.919	$(f \times g) \circ (f' \times g') = (f \circ f') \times (g \circ g')$	$(f \times g) \circ (f' \times g') = (f \circ f') \times (g \circ g').$	$(f \times g) \circ (f' \times g') = (f \circ f') \times (g \circ g').$

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fong-spivak-seven-sketches_3EQ0259	221	■ 0.923	$\begin{aligned} \varphi_{S^{\prime}, T^{\prime}}(\varphi_{S, T}(A, B)) &= \varphi_{S^{\prime}, T^{\prime}}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\}) \\ &= \{(f(a), g(b)) \mid a \in A, b \in B\} \\ &= \operatorname{im}_{f \times g}(A \times B) \\ &= \operatorname{im}_{f \times g}(\varphi_{S, T}(A, B)). \end{aligned}$	$\begin{aligned} \varphi_{S^{\prime}, T^{\prime}}((\operatorname{im}_f \times \operatorname{im}_g)(A \times B)) &= \varphi_{S^{\prime}, T^{\prime}}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\}) \\ &= \{(f(a), g(b)) \mid a \in A, b \in B\} \\ &= \operatorname{im}_{f \times g}(A \times B) \\ &= \operatorname{im}_{f \times g}(\varphi_{S, T}(A, B)). \end{aligned}$	$\begin{aligned} \varphi_{S^{\prime}, T^{\prime}}((\operatorname{im}_f \times \operatorname{im}_g)(A \times B)) &= \varphi_{S^{\prime}, T^{\prime}}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\}) \\ &= \{(f(a), g(b)) \mid a \in A, b \in B\} \\ &= \operatorname{im}_{f \times g}(A \times B) \\ &= \operatorname{im}_{f \times g}(\varphi_{S, T}(A, B)). \end{aligned}$
fong-spivak-seven-sketches_1EQ0087	140	■ 0.928	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}$	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}$	$\mathrm{U}_{\mathcal{P}} \circ \Phi = \Phi = \Phi \circ \mathrm{U}_{\mathcal{Q}}$
fong-spivak-seven-sketches_1EQ0199	288	■ 0.928	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ &\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right) \\ &= (M * (N * P))(w, z). \end{aligned}$	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ &\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right) \\ &= (M * (N * P))(w, z). \end{aligned}$	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ &\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right) \\ &= (M * (N * P))(w, z). \end{aligned}$
fong-spivak-seven-sketches_3EQ0251	313	■ 0.930	$\mathrm{B}(g; (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathrm{B}(g; (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathrm{B}(g \circ (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$
fong-spivak-seven-sketches_1EQ0034	73	■ 0.932	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$
fong-spivak-seven-sketches_9EQ0128	205	■ 0.939	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$
fong-spivak-seven-sketches_1EQ0053 (3.8)	94	■ 0.941	$\mathbf{2} := \operatorname{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \bullet & & \bullet \\ \hline \end{array} \right)$	$\mathbf{2} := \operatorname{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \bullet & & \bullet \\ \hline \end{array} \right)$	$\mathbf{2} := \operatorname{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \bullet & \longrightarrow & \bullet \\ \hline \end{array} \right)$
fong-spivak-seven-sketches_1EQ0075	119	■ 0.943	$\frac{1}{\operatorname{Em}_6}$	<i>(not rendered)</i>	$\frac{1}{\operatorname{Em}_6}$
fong-spivak-seven-sketches_6EQ0026	60	■ 0.948	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$
fong-spivak-seven-sketches_1EQ0111	176	■ 0.950	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$
fong-spivak-seven-sketches_1EQ0159 (7.43)	252	■ 0.951	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\},$	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\},$	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\},$
fong-spivak-seven-sketches_3EQ0007	38	■ 0.952	$f(a) \vee f(b) \not\equiv f(a \vee b).$	$f(a) \vee f(b) \not\equiv f(a \vee b).$	$f(a) \vee f(b) \not\equiv f(a \vee b).$

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fong-spivak-seven-sketches- EQ0030 (2.25)	62	■ 0.954	$\mathrm{H}_2 \mathrm{O} + \mathrm{NaCl} \rightarrow^? \mathrm{H}_2 \mathrm{O}$	$\mathrm{H}_2\mathrm{O} + \mathrm{NaCl} \rightarrow^? \mathrm{H}_2\mathrm{O}$	$\mathrm{H}_2\mathrm{O} + \mathrm{NaCl} \rightarrow^? \mathrm{H}_2\mathrm{O}$
fong-spivak-seven-sketches- EQ0194	287	■ 0.955	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C$	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C.$	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C.$
fong-spivak-seven-sketches- EQ0171 (7.73)	263	■ 0.958	$\forall (t : T) . \exists (s : S) . f(s) = t$	$\forall (t : T) . \exists (s : S) . f(s) = t$	$\forall (t : T) . \exists (s : S) . f(s) = t$
fong-spivak-seven-sketches- EQ0040 (2.80)	81	■ 0.960	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w)$	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w).$	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w).$
fong-spivak-seven-sketches- EQ0262	328	■ 0.963	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) \cap Y$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$
fong-spivak-seven-sketches- EQ0092 (4.40)	143	■ 0.967	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$
fong-spivak-seven-sketches- EQ0218	297	■ 0.967	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in A \text{ and } d_2 \in B\}$	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in A \text{ and } d_2 \in B\}.$	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in A \text{ and } d_2 \in B\}.$
fong-spivak-seven-sketches- EQ0168	260	■ 0.967	$s : S \mid p(s) \vdash q(s) \quad \text{or} \quad p(s) \vdash_{s:S} q(s)$	$s : S \mid p(s) \vdash q(s) \quad \text{or} \quad p(s) \vdash_{s:S} q(s).$	$s : S \mid p(s) \vdash q(s) \quad \text{or} \quad p(s) \vdash_{s:S} q(s).$
fong-spivak-seven-sketches- EQ0066 (3.69)	114	■ 0.968	$f(p) \leq q \text{ if and only if } p \leq g(q)$	$f(p) \leq q \text{ if and only if } p \leq g(q).$	$f(p) \leq q \text{ if and only if } p \leq g(q).$
fong-spivak-seven-sketches- EQ0119 (5.78)	187	■ 0.969	$B_1 ; B_2 := \{(x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2\}$	$B_1; B_2 := \{(x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2\}.$	$B_1 \circ B_2 := \{(x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2\}.$
fong-spivak-seven-sketches- EQ0157	249	■ 0.969	$s_i _{U_i \cap U_j} = s_j _{U_i \cap U_j}$	$s_i _{U_i \cap U_j} = s_j _{U_i \cap U_j}.$	$s_i _{U_i \cap U_j} = s_j _{U_i \cap U_j}.$
fong-spivak-seven-sketches- EQ0100	157	■ 0.973	$\eta_x : \mathbf{1} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$	$\eta_x : \mathbf{1} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V};$	$\eta_x : \mathbf{1} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V};$
fong-spivak-seven-sketches- EQ0045	85	■ 0.974	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v) \quad X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} x(u, v).$	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v) \quad X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} x(u, v).$	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v) \quad \mathcal{X}(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} \mathcal{X}(u, v).$
fong-spivak-seven-sketches- EQ0046	85	■ 0.974	$v \rightarrow w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a$	$v \rightarrow w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a.$	$v \multimap w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a.$
fong-spivak-seven-sketches- EQ0085	139	■ 0.976	$(\Phi ; \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r))$	$(\Phi; \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r)).$	$(\Phi \circ \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r)).$
fong-spivak-seven-sketches- EQ0195	287	■ 0.979	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C$	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C.$	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C.$
fong-spivak-seven-sketches- EQ0173	265	■ 0.979	$\mathbb{R} \cong \{[d, u] \in \mathbb{I}\mathbb{R} \mid d = u\}$	$\mathbb{R} \cong \{[d, u] \in \mathbb{I}\mathbb{R} \mid d = u\}.$	$\mathbb{R} \cong \{[d, u] \in \mathbb{I}\mathbb{R} \mid d = u\}.$
fong-spivak-seven-sketches- EQ0137	220	■ 0.982	$\text{Circ: } (\mathbf{FinSet}, +) \longrightarrow (\mathbf{Set}, \times)$	$\text{Circ: } (\mathbf{FinSet}, +) \longrightarrow (\mathbf{Set}, \times)$	$\text{Circ: } (\mathbf{FinSet}, +) \longrightarrow (\mathbf{Set}, \times)$
fong-spivak-seven-sketches- EQ0169	260	■ 0.983	$\forall (t : T) . p(s, t)$	$\forall (t : T) . p(s, t).$	$\forall (t : T) . p(s, t).$
fong-spivak-seven-sketches- EQ0018	46	■ 0.985	$\operatorname{fix}_j := \{p \in P \mid j(p) \cong p\}$	$\operatorname{fix}_j := \{p \in P \mid j(p) \cong p\}.$	$\operatorname{fix}_j := \{p \in P \mid j(p) \cong p\}.$

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fong-spivak-seven-sketches- EQ0103	165	■ 0.985	$\iota^{\mathrm{\prime}}(x)=\begin{cases}\iota(x) & \text{ if } \iota(x) \in I \\ \iota'(x) & \text{ if } \iota(x) \in \underline{n} \\ \iota'(x) & \text{ if } x \in O.\end{cases}$	$\iota''(x) = \begin{cases} \iota(x) & \text{ if } \iota(x) \in I \\ \iota'(\iota(x)) & \text{ if } \iota(x) \in \underline{n} \\ \iota'(x) & \text{ if } x \in O.\end{cases}$	$\iota''(x) = \begin{cases} \iota(x) & \text{ if } \iota(x) \in I \\ \iota'(\iota(x)) & \text{ if } \iota(x) \in \underline{n} \\ \iota'(x) & \text{ if } x \in O.\end{cases}$
fong-spivak-seven-sketches- EQ0210	292	■ 0.986	$F(a):=a^{\mathrm{\prime}}, \quad G(a):=a^{\mathrm{\prime}}, \quad F(b):=b^{\mathrm{\prime}}, \quad G(b):=b^{\mathrm{\prime}}, \quad F(f):=f_1, \quad \text{ and } G(f):=f_2.$	$F(a) := a', \quad G(a) := a', \quad F(b) := b', \quad G(b) := b', \quad F(f) := f_1, \text{ and } G(f) := f_2.$	$F(a) := a', \quad G(a) := a', \quad F(b) := b', \quad G(b) := b', \quad F(f) := f_1, \text{ and } G(f) := f_2.$
fong-spivak-seven-sketches- EQ0117 (5.75)	186	■ 0.988	$\mathrm{B}(g)=\left\{(x, S(g)(x)) \mid x \in R^m\right\} \subseteq R^m \times R^n$	$B(g)=\{(x, S(g)(x)) \mid x \in R^m\} \subseteq R^m \times R^n.$	$B(g)=\{(x, S(g)(x)) \mid x \in R^m\} \subseteq R^m \times R^n.$
fong-spivak-seven-sketches- EQ0031	67	■ 0.989	$\text{Cost} := ([0, \infty], \geq, 0, +),$	$\text{Cost} := ([0, \infty], \geq, 0, +),$	$\text{Cost} := ([0, \infty], \geq, 0, +),$
fong-spivak-seven-sketches- EQ0204	289	■ 0.989	$\stackrel{\bullet}{v_1} \xrightarrow{f_1} \stackrel{\bullet}{v_2} \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} \stackrel{\bullet}{v_n}$	$\stackrel{\bullet}{v_1} \xrightarrow{f_1} \stackrel{\bullet}{v_2} \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} \stackrel{\bullet}{v_n}$	$\stackrel{\bullet}{v_1} \xrightarrow{f_1} \stackrel{\bullet}{v_2} \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} \stackrel{\bullet}{v_n}$
fong-spivak-seven-sketches- EQ0213	294	■ 0.989	$G:=\stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b} \stackrel{3}{\bullet}$	$G := \stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b} \stackrel{3}{\bullet}$	$G := \left \begin{array}{c} 1 \xrightarrow{a} 2 \xrightarrow{b} 3 \end{array} \right $
fong-spivak-seven-sketches- EQ0250	313	■ 0.990	$\left\{(x, y) \mid \text{there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y)\right\}.$	$\{(x, y) \mid \text{there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y)\}.$	$\{(x, y) \mid \text{there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y)\}.$
fong-spivak-seven-sketches- EQ0038	78	■ 0.990	$\mathcal{X}(x_1, x_2) \leq y(F(x_1), F(x_2)),$	$\mathcal{X}(x_1, x_2) \leq y(F(x_1), F(x_2)),$	$\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2)),$
fong-spivak-seven-sketches- EQ0102	162	■ 0.991	$R; S := \{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}).(i, j) \in R \text{ and } (j, k) \in S\}.$	$R; S := \{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}).(i, j) \in R \text{ and } (j, k) \in S\}.$	$R \circ S := \{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}).(i, j) \in R \text{ and } (j, k) \in S\}.$
fong-spivak-seven-sketches- EQ0080 (4.6)	133	■ 0.991	$\Phi: x^{\mathrm{op}} \times y \rightarrow \mathcal{V}.$	$\Phi: x^{\mathrm{op}} \times y \rightarrow \mathcal{V}.$	$\Phi: \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathcal{V}.$
fong-spivak-seven-sketches- EQ0134	216	■ 0.992	$\varphi_{c_1, c_2}: F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$	$\varphi_{c_1, c_2}: F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$	$\varphi_{c_1, c_2}: F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$
fong-spivak-seven-sketches- EQ0148	236	■ 0.992	$\{p \in P \mid \text{likes_cats}(p)\}.$	<i>(not rendered)</i>	$\{p \in P \mid \text{likes_cats}(p)\}.$
fong-spivak-seven-sketches- EQ0144	228	■ 0.992	$(f \circ \mathrm{id}) := f \circ (\mathrm{id}, \dots, \mathrm{id}, g, \mathrm{id}, \dots, \mathrm{id})$	$(f \circ ig) := f \circ (\mathrm{id}, \dots, \mathrm{id}, g, \mathrm{id}, \dots, \mathrm{id})$	$(f \circ_i g) := f \circ (\mathrm{id}, \dots, \mathrm{id}, g, \mathrm{id}, \dots, \mathrm{id})$
fong-spivak-seven-sketches- EQ0123 (5.86)	189	■ 0.993	$\{(0, (x, x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text{and} \quad \{((x, x), 0) \mid x \in R\} \subseteq R^2 \times R^0.$	$\{(0, (x, x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text{and} \quad \{((x, x), 0) \mid x \in R\} \subseteq R^2 \times R^0.$	$\{(0, (x, x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text{and} \quad \{((x, x), 0) \mid x \in R\} \subseteq R^2 \times R^0.$
fong-spivak-seven-sketches- EQ0229	304	■ 0.993	$x \xrightarrow{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{U_x \times \eta_x} x \times x^{\mathrm{op}} \times x \xrightarrow{\epsilon_x \times U_x} \mathbf{1} \times x \xrightarrow{\alpha} x$	$x \xrightarrow{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{U_x \times \eta_x} x \times x^{\mathrm{op}} \times x \xrightarrow{\epsilon_x \times U_x} \mathbf{1} \times x \xrightarrow{\alpha} x$	$\mathcal{X} \xrightarrow{\alpha^{-1}} \mathcal{X} \times \mathbf{1} \xrightarrow{U_{\mathcal{X}} \times \eta_{\mathcal{X}}} \mathcal{X} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \xrightarrow{\epsilon_{\mathcal{X}} \times U_{\mathcal{X}}} \mathbf{1} \times \mathcal{X} \xrightarrow{\alpha} \mathcal{X}$

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fong-spivak-seven-sketches_1 EQ0225	301	■ 0.993	$\begin{aligned} & ((\Phi \mathrel{\mathbin{\circ}} \Psi) \mathrel{\mathbin{\circ}} \Upsilon)(p, s) \mathrel{\mathbin{\circ}} \\ & = \bigvee_{r \in \mathcal{R}} \left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \right) \otimes \Upsilon(r, s) \\ & = \bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ & = \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}} (\Psi(q, r) \otimes \Upsilon(r, s)) \\ & = (\Phi; (\Psi; \Upsilon))(p, s) \end{aligned}$	$((\Phi; \Psi); \Upsilon)(p, s) = \bigvee_{r \in \mathcal{R}} \left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \right) \otimes \Upsilon(r, s)$ $= \bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s)$ $= \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}} \left(\Psi(q, r) \otimes \Upsilon(r, s) \right)$ $= (\Phi \mathbin{\circ} (\Psi \mathbin{\circ} \Upsilon))(p, s)$	
fong-spivak-seven-sketches_1 EQ0089 (4.29)	140	■ 0.994	$\mathcal{P}\left(p, p_1\right) \otimes \Phi\left(p_1, q\right) \otimes \Phi\left(p_1, q\right) \otimes I \leq \mathcal{P}\left(p, p_1\right) \otimes \Phi\left(p_1, q\right) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q).$	$\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) = \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes I \leq \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q).$	$\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) = \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes I \leq \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q).$
fong-spivak-seven-sketches_1 EQ0165 (7.56)	257	■ 0.995	$(U \wedge V) := U \cap V \quad \text{and} \quad (U \vee V) := U \cup V.$	$(U \wedge V) := U \cap V \quad \text{and} \quad (U \vee V) := U \cup V.$	$(U \wedge V) := U \cap V \quad \text{and} \quad (U \vee V) := U \cup V.$
fong-spivak-seven-sketches_1 EQ0150 (7.14)	241	■ 0.995	$\Omega_{\text{Set}} := \mathbb{B} = \{ \text{true}, \text{false} \}.$	$\Omega_{\text{Set}} := \mathbb{B} = \{ \text{true}, \text{false} \}.$	$\Omega_{\text{Set}} := \mathbb{B} = \{ \text{true}, \text{false} \}.$
fong-spivak-seven-sketches_1 EQ0160 (7.50)	255	■ 0.996	$\Omega(U) := \{ U' \in \mathbf{Op} \mid U' \subseteq U \}.$	$\Omega(U) := \{ U' \in \mathbf{Op} \mid U' \subseteq U \}.$	$\Omega(U) := \{ U' \in \mathbf{Op} \mid U' \subseteq U \}.$
fong-spivak-seven-sketches_1 EQ0260	324	■ 0.996	$p \mathrel{\mathbin{\circ}} h_3 = r \mathrel{\mathbin{\circ}} g \mathrel{\mathbin{\circ}} h_3 = r \mathrel{\mathbin{\circ}} h_2 \mathrel{\mathbin{\circ}} g' = q \mathrel{\mathbin{\circ}} f' \mathrel{\mathbin{\circ}} g'$	$p; h_3 = r; g; h_3 = r; h_2; g' = q; f'; g'$	$p \mathbin{\circ} h_3 = r \mathbin{\circ} g \mathbin{\circ} h_3 = r \mathbin{\circ} h_2 \mathbin{\circ} g' = q \mathbin{\circ} f' \mathbin{\circ} g'$
fong-spivak-seven-sketches_1 EQ0088 (4.28)	140	■ 0.996	$\Phi(p, q) = I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \leq \bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) = (U_P; \Phi)(p, q).$	$\Phi(p, q) = I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \leq \bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) = (U_P; \Phi)(p, q).$	$\Phi(p, q) = I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \leq \bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) = (U_P \mathbin{\circ} \Phi)(p, q).$
fong-spivak-seven-sketches_1 EQ0185	276	■ 0.996	$f^*(P) = \{ \{ x \in \mathbb{Z} \mid x \leq 0 \}, \{ x \in \mathbb{Z} \mid x \geq 1 \} \} \quad \text{and} \quad f^*(Q) = \{ \{ 0 \}, \{ x \in \mathbb{Z} \mid x \neq 0 \} \}.$	$f^*(P) = \{ \{ x \in \mathbb{Z} \mid x \leq 0 \}, \{ x \in \mathbb{Z} \mid x \geq 1 \} \} \quad \text{and} \quad f^*(Q) = \{ \{ 0 \}, \{ x \in \mathbb{Z} \mid x \neq 0 \} \}.$	$f^*(P) = \{ \{ x \in \mathbb{Z} \mid x \leq 0 \}, \{ x \in \mathbb{Z} \mid x \geq 1 \} \} \quad \text{and} \quad f^*(Q) = \{ \{ 0 \}, \{ x \in \mathbb{Z} \mid x \neq 0 \} \}.$
fong-spivak-seven-sketches_1 EQ0166 (7.57)	257	■ 0.996	$(U \Rightarrow V) := \bigcup_{\{ R \in \mathbf{Op} \mid R \cap U \subseteq V \}} R.$	$(U \Rightarrow V) := \bigcup_{\{ R \in \mathbf{Op} \mid R \cap U \subseteq V \}} R.$	$(U \Rightarrow V) := \bigcup_{\{ R \in \mathbf{Op} \mid R \cap U \subseteq V \}} R.$
fong-spivak-seven-sketches_1 EQ0013 (1.116)	44	■ 0.997	$f(p) := \bigwedge \{ q \in Q \mid p \leq g(q) \};$	$f(p) := \bigwedge \{ q \in Q \mid p \leq g(q) \};$	$f(p) := \bigwedge \{ q \in Q \mid p \leq g(q) \};$
fong-spivak-seven-sketches_1 EQ0154	243	■ 0.997	$\begin{array}{cc c c} P & Q & P \wedge Q & P = (P \wedge Q) \\ \hline \text{true} & \text{true} & ? & ? \\ \text{true} & \text{false} & ? & ? \\ \text{false} & \text{true} & ? & ? \\ \text{false} & \text{false} & ? & ? \end{array}$	$\begin{array}{cc c c} P & Q & P \wedge Q & P = (P \wedge Q) \\ \hline \text{true} & \text{true} & ? & ? \\ \text{true} & \text{false} & ? & ? \\ \text{false} & \text{true} & ? & ? \\ \text{false} & \text{false} & ? & ? \end{array}$	$\begin{array}{cc c c} P & Q & P \wedge Q & P = (P \wedge Q) \\ \hline \text{true} & \text{true} & ? & ? \\ \text{true} & \text{false} & ? & ? \\ \text{false} & \text{true} & ? & ? \\ \text{false} & \text{false} & ? & ? \end{array}$

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fong-spivak-seven-sketches_1 EQ0041	81	■ 0.998	$a+x \geq y \quad \text{\texttt{\text{iff}}} \quad a \geq y-x \quad \text{\texttt{\text{iff}}} \quad \max(0, a) \geq \max(0, y-x) \quad \text{\texttt{\text{iff}}} \quad a \geq (x \multimap y)$	$a+x \geq y \quad \text{\texttt{\text{iff}}} \quad a \geq y-x \quad \text{\texttt{\text{iff}}} \quad \max(0, a) \geq \max(0, y-x) \quad \text{\texttt{\text{iff}}} \quad a \geq (x \multimap y)$	$a+x \geq y \quad \text{\texttt{\text{iff}}} \quad a \geq y-x \quad \text{\texttt{\text{iff}}} \quad \max(0, a) \geq \max(0, y-x) \quad \text{\texttt{\text{iff}}} \quad a \geq (x \multimap y)$
fong-spivak-seven-sketches_2 EQ0082 (4.14)	135	■ 0.998	$\Phi(B, x)=11, \quad \Phi(A, z)=20, \quad \Phi(C, y)=17 \text{ .}$	$\Phi(B, x)=11, \quad \Phi(A, z)=20, \quad \Phi(C, y)=17.$	$\Phi(B, x)=11, \quad \Phi(A, z)=20, \quad \Phi(C, y)=17.$
fong-spivak-seven-sketches_3 EQ0124	190	■ 0.998	$v=\int \frac{1}{m} F(t) \, dt + u(t) + p \int a u(t) + b v(t) \, dt \text{ .}$	$v = \int \frac{1}{m} F(t) dt + u(t) + p \int a u(t) + b v(t) dt.$	$v = \int \frac{1}{m} F(t) dt + u(t) + p \int a u(t) + b v(t) dt.$
fong-spivak-seven-sketches_4 EQ0198	288	■ 0.998	$\begin{aligned} I_X * M(x, y) &= \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y) \\ &= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right) \\ &= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right) \\ &= M(x, y) \vee 0 = M(x, y). \end{aligned}$	$\begin{aligned} I_X * M(x, y) &= \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y) \\ &= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right) \\ &= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right) \\ &= M(x, y) \vee 0 = M(x, y). \end{aligned}$	$\begin{aligned} I_X * M(x, y) &= \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y) \\ &= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right) \\ &= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right) \\ &= M(x, y) \vee 0 = M(x, y). \end{aligned}$
fong-spivak-seven-sketches_5 EQ0070	117	■ 0.999	$\Sigma_F(I)(\text{\texttt{Airline Seat}})=I(\text{\texttt{Economy}}) \sqcup I(\text{\texttt{First Class}}) \text{ .}$	$\Sigma_F(I)(\text{\texttt{Airline Seat}}) = I(\text{\texttt{Economy}}) \sqcup I(\text{\texttt{First Class}}) \text{ .}$	$\Sigma_F(I)(\text{\texttt{Airline Seat}}) = I(\text{\texttt{Economy}}) \sqcup I(\text{\texttt{First Class}}).$
fong-spivak-seven-sketches_6 EQ0084 (4.20)	138	■ 0.999	$(\Phi \mathbin{;} \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r) \text{ .}$	$(\Phi; \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r).$	$(\Phi \mathbin{;} \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r).$
fong-spivak-seven-sketches_7 EQ0008 (1.96)	39	■ 0.999	$f(p) \leq q \quad \text{\texttt{\text{if and only if}}} \quad p \leq g(q) \text{ .}$	$f(p) \leq q \quad \text{\texttt{\text{if and only if}}} \quad p \leq g(q).$	$f(p) \leq q \quad \text{\texttt{\text{if and only if}}} \quad p \leq g(q).$
fong-spivak-seven-sketches_8 EQ0035	76	■ 0.999	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone }) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f) \end{array}$	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone }) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f) \end{array}$	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone }) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f) \end{array}$
fong-spivak-seven-sketches_9 EQ0042	82	■ 0.999	$a+c \rightarrow d \quad \text{\texttt{\text{if and only if}}} \quad a \rightarrow (c \rightarrow d) \text{ .}$	$a+c \rightarrow d \quad \text{\texttt{\text{if and only if}}} \quad a \rightarrow (c \rightarrow d).$	$a+c \rightarrow d \quad \text{\texttt{\text{if and only if}}} \quad a \rightarrow (c \multimap d).$
fong-spivak-seven-sketches_10 EQ0065 (3.66)	113	■ 0.999	$\text{\texttt{Gr}} \xrightarrow{F} \text{\texttt{DDS}} \xrightarrow{I} \text{\texttt{Set}} \text{ .}$	$\text{\texttt{Gr}} \xrightarrow{F} \text{\texttt{DDS}} \xrightarrow{I} \text{\texttt{Set}}.$	$\text{\texttt{Gr}} \xrightarrow{F} \text{\texttt{DDS}} \xrightarrow{I} \text{\texttt{Set}}.$
fong-spivak-seven-sketches_11 EQ0126 (6.15)	199	■ 0.999	$[f, g](x) = \begin{cases} f(x) & \text{\texttt{\text{if}}} \ x = \iota_A(a) \text{\texttt{\text{ for some}}} \ a \in A; \\ g(x) & \text{\texttt{\text{if}}} \ x = \iota_B(b) \text{\texttt{\text{ for some}}} \ b \in B \text{ .} \end{cases}$	$[f, g](x) = \begin{cases} f(x) & \text{\texttt{\text{if}}} \ x = \iota_A(a) \text{\texttt{\text{ for some}}} \ a \in A; \\ g(x) & \text{\texttt{\text{if}}} \ x = \iota_B(b) \text{\texttt{\text{ for some}}} \ b \in B. \end{cases}$	$[f, g](x) = \begin{cases} f(x) & \text{\texttt{\text{if}}} \ x = \iota_A(a) \text{\texttt{\text{ for some}}} \ a \in A; \\ g(x) & \text{\texttt{\text{if}}} \ x = \iota_B(b) \text{\texttt{\text{ for some}}} \ b \in B. \end{cases}$

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fong-spivak-seven-sketches EQ0139	220	■ 0.999	$\begin{aligned} \operatorname{Circ}(f): \operatorname{Circ}(V) &\longrightarrow \operatorname{Circ}(V') \\ (V, A, s, t, \ell) &\longmapsto (V', A, (s; f), (t; f), \ell). \end{aligned}$	$\operatorname{Circ}(f) : \operatorname{Circ}(V) \longrightarrow \operatorname{Circ}(V'); \\ (V, A, s, t, \ell) \longmapsto (V', A, (s; f), (t; f), \ell).$	$\operatorname{Circ}(f): \operatorname{Circ}(V) \longrightarrow \operatorname{Circ}(V'); \\ (V, A, s, t, \ell) \longmapsto (V', A, (s \circ f), (t \circ f), \ell).$
fong-spivak-seven-sketches EQ0032	68	■ 1.000	$d(x) := \begin{cases} \text{false} & \text{if } x > 0 \\ \text{true} & \text{if } x = 0 \end{cases} \quad u(x) := \begin{cases} \text{false} & \text{if } x = \infty \\ \text{true} & \text{if } x < \infty \end{cases}$	$d(x) := \begin{cases} \text{false} & \text{if } x > 0 \\ \text{true} & \text{if } x = 0 \end{cases} \quad u(x) := \begin{cases} \text{false} & \text{if } x = \infty \\ \text{true} & \text{if } x < \infty \end{cases}$	$d(x) := \begin{cases} \text{false} & \text{if } x > 0 \\ \text{true} & \text{if } x = 0 \end{cases} \quad u(x) := \begin{cases} \text{false} & \text{if } x = \infty \\ \text{true} & \text{if } x < \infty \end{cases}$
fong-spivak-seven-sketches EQ0176	272	■ 1.000	$\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$	$\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$	$\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$
fong-spivak-seven-sketches EQ0252	313	■ 1.000	$\begin{aligned} \operatorname{B}(g^{\operatorname{op}}) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\} \\ \operatorname{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\} \end{aligned}$	$\begin{aligned} \operatorname{B}(g^{\operatorname{op}}) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\} \\ \operatorname{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\} \end{aligned}$	$\begin{aligned} \operatorname{B}(g^{\operatorname{op}}) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\} \\ \operatorname{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\} \end{aligned}$
fong-spivak-seven-sketches EQ0001 (1.8)	18	■ 1.000	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$
fong-spivak-seven-sketches EQ0004	24	■ 1.000	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d$	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d$	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d$
fong-spivak-seven-sketches EQ0006	35	■ 1.000	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$
fong-spivak-seven-sketches EQ0009	39	■ 1.000	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$
fong-spivak-seven-sketches EQ0011	43	■ 1.000	$g(\bigwedge A) \cong \bigwedge g(A).$	$g(\bigwedge A) \cong \bigwedge g(A).$	$g\left(\bigwedge A\right) \cong \bigwedge g(A).$
fong-spivak-seven-sketches EQ0012	43	■ 1.000	$f(\bigvee A) \cong \bigvee f(A)$	$f(\bigvee A) \cong \bigvee f(A)$	$f\left(\bigvee A\right) \cong \bigvee f(A).$
fong-spivak-seven-sketches EQ0014	44	■ 1.000	$p_{\{0\}} \leq \bigwedge \{g(q) \in P \mid p_{\{0\}} \leq g(q)\} \cong g\left(\bigwedge \{q \in Q \mid p_{\{0\}} \leq g(q)\}\right) = g(f(p_0)),$	$p_0 \leq \bigwedge \{g(q) \in P \mid p_0 \leq g(q)\} \cong g\left(\bigwedge \{q \in Q \mid p_0 \leq g(q)\}\right) = g(f(p_0)),$	$p_0 \leq \bigwedge \{g(q) \in P \mid p_0 \leq g(q)\} \cong g\left(\bigwedge \{q \in Q \mid p_0 \leq g(q)\}\right) = g(f(p_0)),$
fong-spivak-seven-sketches EQ0015	44	■ 1.000	$f\left(g\left(\bigwedge_{q \in Q} q\right)\right) = \bigwedge_{q \in Q} f(g(q)) \leq \bigwedge_{q \in Q} q = q_0,$	$f(g(q_0)) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$	$f(g(q_0)) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$
fong-spivak-seven-sketches EQ0016	45	■ 1.000	$f_{\dagger}(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}$	$f_{\dagger}(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}$	$f_{\dagger}(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}$
fong-spivak-seven-sketches EQ0017	45	■ 1.000	$f_{*}(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}$	$f_{*}(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}$	$f_{*}(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}$
fong-spivak-seven-sketches EQ0019 (2.7)	54	■ 1.000	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$

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fong-spivak-seven-sketches_59 EQ0023 (2.16)	59	■ 1.000	$t \leq v+w \quad w+u \leq x+z \quad v+x \leq y$	$t \leq v+w \quad w+u \leq x+z \quad v+x \leq y$	$t \leq v+w \quad w+u \leq x+z \quad v+x \leq y$
fong-spivak-seven-sketches_59 EQ0024 (2.17)	59	■ 1.000	$t+u \leq y+z$.	$t+u \leq y+z$.	$t+u \leq y+z$.
fong-spivak-seven-sketches_59 EQ0025 (2.18)	59	■ 1.000	$t+u \leq v+w+u \leq v+x+z \leq y+z$.	$t+u \leq v+w+u \leq v+x+z \leq y+z$.	$t+u \leq v+w+u \leq v+x+z \leq y+z$.
fong-spivak-seven-sketches_61 EQ0027 (2.24)	61	■ 1.000	$2 \mathrm{H}_{\mathrm{2}} \mathrm{Na} \rightarrow 2 \mathrm{NaOH}+\mathrm{H}_{\mathrm{2}}$.	$2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na} \rightarrow 2\mathrm{NaOH} + \mathrm{H}_2$.	$2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na} \rightarrow 2\mathrm{NaOH} + \mathrm{H}_2$.
fong-spivak-seven-sketches_61 EQ0028 (2.22)	61	■ 1.000	$y+k \rightarrow y^{\prime}+k^{\prime} \quad x+y^{\prime} \rightarrow z^{\prime} \quad z^{\prime}+k^{\prime} \rightarrow z+k$ $\rightarrow z^{\prime} \quad z^{\prime}+k^{\prime} \rightarrow z+k$ $\rightarrow z+k$	$y+k \rightarrow y'+k' \quad x+y' \rightarrow z' \quad z'+k' \rightarrow z+k$	$y+k \rightarrow y'+k' \quad x+y' \rightarrow z' \quad z'+k' \rightarrow z+k$
fong-spivak-seven-sketches_61 EQ0029 (2.23)	61	■ 1.000	$x+y+k \rightarrow x+y^{\prime}+k^{\prime} \rightarrow z^{\prime}+k^{\prime} \rightarrow z+k$.	$x+y+k \rightarrow x+y'+k' \rightarrow z'+k' \rightarrow z+k$.	$x+y+k \rightarrow x+y'+k' \rightarrow z'+k' \rightarrow z+k$.
fong-spivak-seven-sketches_77 EQ0037 (2.66)	77	■ 1.000	$f(x):= \begin{cases} \text{true} & \text{if } x=0 \\ \text{false} & \text{if } x>0 \end{cases}$	$f(x) := \begin{cases} \text{true} & \text{if } x = 0 \\ \text{false} & \text{if } x > 0 \end{cases}$	$f(x) := \begin{cases} \text{true} & \text{if } x = 0 \\ \text{false} & \text{if } x > 0 \end{cases}$
fong-spivak-seven-sketches_82 EQ0044 (2.88)	82	■ 1.000	$\left(v \otimes \bigvee_{a \in A} a\right) \cong \bigvee_{a \in A}\left(v \otimes a\right)$.	$\left(v \otimes \bigvee_{a \in A} a\right) \cong \bigvee_{a \in A}\left(v \otimes a\right)$.	$\left(v \otimes \bigvee_{a \in A} a\right) \cong \bigvee_{a \in A}\left(v \otimes a\right)$.
fong-spivak-seven-sketches_85 EQ0047 (2.99)	85	■ 1.000	$(M * N)(i, k)=\sum_{j} M(i, j) * N(j, k)$.	$(M * N)(i, k)=\sum_j M(i, j) * N(j, k)$.	$(M * N)(i, k)=\sum_j M(i, j) * N(j, k)$.
fong-spivak-seven-sketches_86 EQ0048 (2.101)	86	■ 1.000	$(M * N)(x, z):=\bigvee_{y \in Y} M(x, y) \otimes N(y, z)$.	$(M * N)(x, z):=\bigvee_{y \in Y} M(x, y) \otimes N(y, z)$.	$(M * N)(x, z):=\bigvee_{y \in Y} M(x, y) \otimes N(y, z)$.
fong-spivak-seven-sketches_86 EQ0049	86	■ 1.000	$\left(\begin{array}{cc} \text{false} & \text{false} \\ \text{false} & \text{true} \\ \text{true} & \text{true} \end{array}\right) *\left(\begin{array}{ccc} \text{true} & \text{true} & \text{false} \\ \text{true} & \text{false} & \text{true} \end{array}\right)=\left(\begin{array}{ccc} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{array}\right)$	$\left(\begin{array}{ccc} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{array}\right)$	$\left(\begin{array}{ccc} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{array}\right)$
fong-spivak-seven-sketches_86 EQ0050	86	■ 1.000	$I_X(x, y):= \begin{cases} I & \text{if } x=y \\ \text{false} & \text{if } x \neq y \end{cases}$	$I_X(x, y):= \begin{cases} I & \text{if } x = y \\ 0 & \text{if } x \neq y. \end{cases}$	$I_X(x, y):= \begin{cases} I & \text{if } x = y \\ 0 & \text{if } x \neq y. \end{cases}$
fong-spivak-seven-sketches_87 EQ0051	87	■ 1.000	$M_{\mathrm{Y}}^2=\begin{array}{c ccc} & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$	$M_Y^2 = \begin{array}{c ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$	$M_Y^2 = \begin{array}{c ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$

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fong-spivak-seven-sketches_1EQ0052	87	■ 1.000	$M_{\{Y\}^{\{3\}}}=\begin{array}{c ccc}\nearrow&x&y&z\\\hline x&0&4&3\\ y&3&0&6\\ z&7&4&0\\\end{array}$	$M_Y^3 = \begin{array}{c ccc}\nearrow&x&y&z\\\hline x&0&4&3\\ y&3&0&6\\ z&7&4&0\\\end{array}$	$M_Y^3 = \begin{array}{c ccc}\nearrow&x&y&z\\\hline x&0&4&3\\ y&3&0&6\\ z&7&4&0\\\end{array}$
fong-spivak-seven-sketches_1EQ0055	96	■ 1.000	$\{A, B, C, D, f, g, h, i, f ; h\}$	$\{A, B, C, D, f, g, h, i, f; h\}$	$\{A, B, C, D, f, g, h, i, f \circ h\}$
fong-spivak-seven-sketches_1EQ0058	104	■ 1.000	$f^{\{ \prime \}} ; h^{\{ \prime \}}=F(f) ; F(h)=F(f ; h)=F(g ; i)=F(g) ; F(i)=g^{\{ \prime \}} ; i^{\{ \prime \}}$	$f';h'=F(f);F(h)=F(f;h)=F(g;i)=F(g);F(i)=g';i'$	$f' \circ h' = F(f) \circ F(h) = F(f \circ h) = F(g \circ i) = F(g) \circ F(i) = g' \circ i'$
fong-spivak-seven-sketches_1EQ0060 (3.50)	107	■ 1.000	$F(f) ; \alpha_d=\alpha_c ; G(f) .$	$F(f); \alpha_d = \alpha_c; G(f).$	$F(f) \circ \alpha_d = \alpha_c \circ G(f).$
fong-spivak-seven-sketches_1EQ0062	110	■ 1.000	$\alpha_{\text {Vertex }}: G(\text {Vertex }) \rightarrow H(\text {Vertex }) \quad \text {and} \quad \alpha_{\text {Arrow }}: G(\text {Arrow }) \rightarrow H(\text {Arrow }),$	$\alpha_{\text{Vertex}} : G(\text{Vertex}) \rightarrow H(\text{Vertex}) \quad \text{and} \quad \alpha_{\text{Arrow}} : G(\text{Arrow}) \rightarrow H(\text{Arrow}),$	$\alpha_{\text{Vertex}}: G(\text{Vertex}) \rightarrow H(\text{Vertex}) \quad \text{and} \quad \alpha_{\text{Arrow}}: G(\text{Arrow}) \rightarrow H(\text{Arrow}),$
fong-spivak-seven-sketches_1EQ0067	114	■ 1.000	$\alpha_{c, d}: \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$	$\alpha_{c,d}: \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$	$\alpha_{c,d}: \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$
fong-spivak-seven-sketches_1EQ0076	121	■ 1.000	$X \times Y:=\{(x, y) \mid x \in X, y \in Y\},$	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$
fong-spivak-seven-sketches_1EQ0090	141	■ 1.000	$(\Phi ; \Psi) ; \Upsilon=\Phi ;(\Psi ; \Upsilon) .$	$(\Phi; \Psi); \Upsilon = \Phi; (\Psi; \Upsilon).$	$(\Phi \circ \Psi) \circ \Upsilon = \Phi \circ (\Psi \circ \Upsilon).$
fong-spivak-seven-sketches_1EQ0095	148	■ 1.000	$\alpha_{s, t, u}: \{(s,(t, u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s, t), u) \mid s \in S, t \in T, u \in U\} .$	$\alpha_{s,t,u}: \{(s,(t,u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s,t),u) \mid s \in S, t \in T, u \in U\}.$	$\alpha_{s,t,u}: \{(s,(t,u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s,t),u) \mid s \in S, t \in T, u \in U\}.$
fong-spivak-seven-sketches_1EQ0099	156	■ 1.000	$(\Phi \times \Psi)\left(\left(x_1, y_1\right),\left(x_2, y_2\right)\right):=\Phi\left(x_1, x_2\right) \otimes \Psi\left(y_1, y_2\right) .$	$(\Phi \times \Psi)\left(\left(x_1, y_1\right),\left(x_2, y_2\right)\right):=\Phi\left(x_1, x_2\right) \otimes \Psi\left(y_1, y_2\right) .$	$(\Phi \times \Psi)\left(\left(x_1, y_1\right),\left(x_2, y_2\right)\right):=\Phi\left(x_1, x_2\right) \otimes \Psi\left(y_1, y_2\right).$
fong-spivak-seven-sketches_1EQ0101 (5.4)	162	■ 1.000	$i \longmapsto \begin{cases} f(i) & \text { if } 1 \leq i \leq m \\ m^{\prime}+g(i) & \text { if } m+1 \leq i \leq m+n \end{cases}$	$i \mapsto \begin{cases} f(i) & \text { if } 1 \leq i \leq m; \\ m' + g(i) & \text { if } m + 1 \leq i \leq m + n. \end{cases}$	$i \mapsto \begin{cases} f(i) & \text { if } 1 \leq i \leq m; \\ m' + g(i) & \text { if } m + 1 \leq i \leq m + n. \end{cases}$
fong-spivak-seven-sketches_1EQ0106 (5.26)	168	■ 1.000	$G:=\left\{\rho_{m, n} \mid m, n \in \mathbb{N}\right\}, \quad \quad s\left(\rho_{m, n}\right):=m, \quad t\left(\rho_{m, n}\right):=n,$	$G:=\left\{\rho_{m, n} \mid m, n \in \mathbb{N}\right\}, \quad s\left(\rho_{m, n}\right):=m, \quad t\left(\rho_{m, n}\right):=n,$	$G:=\left\{\rho_{m, n} \mid m, n \in \mathbb{N}\right\}, \quad s\left(\rho_{m, n}\right):=m, \quad t\left(\rho_{m, n}\right):=n,$
fong-spivak-seven-sketches_1EQ0108	174	■ 1.000	$\begin{array}{l} D \ x+3 D^2 y-2 z=0 \\ D^2 y+5 D z=0 \end{array}$	$\begin{array}{l} D x+3 D^2 y-2 z=0 \\ D^2 y+5 D z=0 \end{array}$	$\begin{array}{l} D x+3 D^2 y-2 z=0 \\ D^2 y+5 D z=0 \end{array}$
fong-spivak-seven-sketches_1EQ0110 (5.48)	176	■ 1.000	$M ; N(a, c):=\sum_{b \in \underline{n}} M(a, b) \times N(b, c),$	$M; N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c),$	$M \circ N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c),$
fong-spivak-seven-sketches_1EQ0112	176	■ 1.000	$A=\left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array}\right) \quad B=\left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array}\right) .$	$A=\left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array}\right) \quad B=\left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array}\right) .$	$A=\left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array}\right) \quad B=\left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array}\right) .$
fong-spivak-seven-sketches_1EQ0114	179	■ 1.000	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k) .$	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k).$	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k).$

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fong-spivak-seven-sketches- EQ0118 (5.76)	186	■ 1.000	$\mathrm{B}\backslash\left(g^{\{\mathrm{op}\}}\backslash\right):=\backslash\left\{\left(S(g)(x), x\right) \mid x \in R^{\{m\}}\backslash\right\} \backslash\subseteq R^{\{n\}} \backslash\times R^{\{m\}} \backslash.$	$B\left(g^{\mathrm{op}}\right):=\left\{\left(S(g)(x), x\right) \mid x \in R^m\right\} \subseteq R^n \times R^m.$	$B\left(g^{\mathrm{op}}\right):=\left\{\left(S(g)(x), x\right) \mid x \in R^m\right\} \subseteq R^n \times R^m.$
fong-spivak-seven-sketches- EQ0122	188	■ 1.000	$\mathrm{B}\backslash\left(\backslash\left(g^{\{\mathrm{op}\}}\backslash\right) ; h\backslash\right)=\backslash\left\{\left(S(g)(x), S(h)(x)\right) \mid x \in R^{\{m\}}\backslash\right\} \backslash.$	$B\left(\left(g^{\mathrm{op}}\right) ; h\right)=\left\{\left(S(g)(x), S(h)(x)\right) \mid x \in R^m\right\}.$	$B\left(\left(g^{\mathrm{op}}\right) ; h\right)=\left\{\left(S(g)(x), S(h)(x)\right) \mid x \in R^m\right\}.$
fong-spivak-seven-sketches- EQ0127	203	■ 1.000	$y(0)=y(g(0))=x(f(0))=x(0) \backslash.$	$y(0)=y(g(0))=x(f(0))=x(0).$	$y(0)=y(g(0))=x(f(0))=x(0).$
fong-spivak-seven-sketches- EQ0130	206	■ 1.000	$\{(v, d) \mid d \in D(v)\}$	$\{(v, d) \mid d \in D(v)\}$	$\{(v, d) \mid d \in D(v)\}$
fong-spivak-seven-sketches- EQ0135 (6.76)	219	■ 1.000	$F\backslash\left(\backslash\left[\iota_{\{N\}}, \iota_{\{P\}}\backslash\right]\backslash\right)\backslash\left(\backslash\mathrm{varphi}_{\{N, P\}}(s, t)\backslash\right)$	$F\left([\iota_N, \iota_P]\right)\left(\varphi_{N, P}(s, t)\right)$	$F\left([\iota_N, \iota_P]\right)\left(\varphi_{N, P}(s, t)\right)$
fong-spivak-seven-sketches- EQ0138	220	■ 1.000	$\mathrm{operatorname{Circ}}(V):=\{(V, A, s, t, \ell) \mid \text { where } s, t: A \rightarrow V, \ell: E \rightarrow C\} \backslash.$	$\mathrm{Circ}(V):=\{(V, A, s, t, \ell) \mid \text { where } s, t: A \rightarrow V, \ell: E \rightarrow C\}.$	$\mathrm{Circ}(V):=\{(V, A, s, t, \ell) \mid \text { where } s, t: A \rightarrow V, \ell: E \rightarrow C\}.$
fong-spivak-seven-sketches- EQ0141	225	■ 1.000	$\mathrm{circ}_{\{i\}}: \mathcal{O}\backslash\left(s_{\{1\}}, \backslash\mathrm{ldots}, s_{\{m\}} ; t_{\{i\}}\backslash\right) \backslash\times \mathcal{O}\backslash\left(t_{\{1\}}, \backslash\mathrm{ldots}, t_{\{n\}} ; t\backslash\right) \backslash\rightarrow \mathcal{O}\backslash\left(t_{\{1\}}, \backslash\mathrm{ldots}, t_{\{i-1\}}, s_{\{1\}}, \backslash\mathrm{ldots}, s_{\{m\}}, t_{\{i+1\}}, \backslash\mathrm{ldots}, t_{\{n\}} ; t\backslash\right) ;$	$\circ_i: \mathcal{O}\left(s_1, \ldots, s_m ; t_i\right) \times \mathcal{O}\left(t_1, \ldots, t_n ; t\right) \rightarrow \mathcal{O}\left(t_1, \ldots, t_{i-1}, s_1, \ldots, s_m, t_{i+1}, \ldots, t_n ; t\right) ;$	$\circ_i: \mathcal{O}\left(s_1, \ldots, s_m ; t_i\right) \times \mathcal{O}\left(t_1, \ldots, t_n ; t\right) \rightarrow \mathcal{O}\left(t_1, \ldots, t_{i-1}, s_1, \ldots, s_m, t_{i+1}, \ldots, t_n ; t\right) ;$
fong-spivak-seven-sketches- EQ0145	229	■ 1.000	$F: \mathcal{O}\backslash\left(t_{\{1\}}, \backslash\mathrm{ldots}, t_{\{n\}} ; t\backslash\right) \backslash\rightarrow \mathcal{P}\backslash\left(f\backslash\left(t_{\{1\}}\backslash\right), \backslash\mathrm{ldots}, f\backslash\left(t_{\{n\}}\backslash\right) ; f(t)\backslash\right)$	$F: \mathcal{O}\left(t_1, \ldots, t_n ; t\right) \rightarrow \mathcal{P}\left(f\left(t_1\right), \ldots, f\left(t_n\right) ; f(t)\right)$	$F: \mathcal{O}\left(t_1, \ldots, t_n ; t\right) \rightarrow \mathcal{P}\left(f\left(t_1\right), \ldots, f\left(t_n\right) ; f(t)\right)$
fong-spivak-seven-sketches- EQ0149 (7.10)	239	■ 1.000	$\mathcal{C}\{A \times C, D\} \cong \mathcal{C}\{A, D^{\{C\}}\}$	$\mathcal{C}\left(A \times C, D\right) \cong \mathcal{C}\left(A, D^C\right)$	$\mathcal{C}\left(A \times C, D\right) \cong \mathcal{C}\left(A, D^C\right)$
fong-spivak-seven-sketches- EQ0151	241	■ 1.000	$\ulcorner m\urcorner(y):=\begin{cases} \text { true } & \& \text { if } m(x)=y \text { for some } x \in X \\ \text { false } & \& \text { otherwise } \end{cases}$	$\ulcorner m\urcorner(y):=\begin{cases} \text { true } & \text { if } m(x)=y \text { for some } x \in X \\ \text { false } & \text { otherwise } \end{cases}$	$\ulcorner m\urcorner(y):=\begin{cases} \text { true } & \text { if } m(x)=y \text { for some } x \in X \\ \text { false } & \text { otherwise } \end{cases}$
fong-spivak-seven-sketches- EQ0152 (7.15)	241	■ 1.000	$X=\{y \in Y \mid p(y)=\text { true }\} \backslash.$	$X=\{y \in Y \mid p(y)=\text { true }\}.$	$X=\{y \in Y \mid p(y)=\text { true }\}.$
fong-spivak-seven-sketches- EQ0155	246	■ 1.000	$B(p ; \epsilon):=\left\{p^{\{\prime\}} \in \mathbb{R}^{\{2\}} \mid d\left(p, p^{\{\prime\}}\right)<\epsilon\right\}^7$	$B(p ; \epsilon):=\left\{p^{\prime} \in \mathbb{R}^2 \mid d\left(p, p^{\prime}\right)<\epsilon\right\}^7$	$B(p ; \epsilon):=\left\{p^{\prime} \in \mathbb{R}^2 \mid d\left(p, p^{\prime}\right)<\epsilon\right\}^7$
fong-spivak-seven-sketches- EQ0161 (7.51)	255	■ 1.000	$\begin{aligned} & \Omega(U) \& \rightarrow \Omega(V) \backslash \backslash U^{\{\prime\}} \\ & U^{\prime} \mapsto U^{\prime} \cap V . \end{aligned}$	$\Omega(U) \rightarrow \Omega(V)$ $U^{\prime} \mapsto U^{\prime} \cap V.$	$\Omega(U) \rightarrow \Omega(V)$ $U^{\prime} \mapsto U^{\prime} \cap V.$
fong-spivak-seven-sketches- EQ0162	256	■ 1.000	$V \cap U_{\{j\}}=\left(\bigcup_{i \in I} V_{\{i\}}\right) \cap U_{\{j\}}=\bigcup_{i \in I}\left(V_{\{i\}} \cap U_{\{j\}}\right)=\bigcup_{i \in I}\left(V_{\{j\}} \cap U_{\{i\}}\right)=\left(\bigcup_{i \in I} U_{\{i\}}\right) \cap V_{\{j\}}=V_{\{j\}}$	$V \cap U_j=\left(\bigcup_{i \in I} V_i\right) \cap U_j=\bigcup_{i \in I}\left(V_i \cap U_j\right)=\bigcup_{i \in I}\left(V_j \cap U_i\right)=\left(\bigcup_{i \in I} U_i\right) \cap V_j=V_j$	$V \cap U_j=\left(\bigcup_{i \in I} V_i\right) \cap U_j=\bigcup_{i \in I}\left(V_i \cap U_j\right)=\bigcup_{i \in I}\left(V_j \cap U_i\right)=\left(\bigcup_{i \in I} U_i\right) \cap V_j=V_j$
fong-spivak-seven-sketches- EQ0167 (7.63)	259	■ 1.000	$\left \left(\Omega^{\{S\}}\backslash\right)\right , \leq^{\{S\}} \backslash\right)$	$\left(\left \Omega^S\right , \leq^S\right)$	$\left(\left \Omega^S\right , \leq^S\right)$
fong-spivak-seven-sketches- EQ0170	261	■ 1.000	$p(n, z)=\begin{cases} \text { true } & \& \text { if } n \leq z \\ \text { false } & \& \text { if } n> z . \end{cases}$	$p(n, z)=\begin{cases} \text { true } & \text { if } n \leq z \\ \text { false } & \text { if } n> z . \end{cases}$	$p(n, z)=\begin{cases} \text { true } & \text { if } n \leq z \\ \text { false } & \text { if } n> z . \end{cases}$
fong-spivak-seven-sketches- EQ0174	265	■ 1.000	$S\backslash\left(o_{\{[a, b]\}}\backslash\right) \cong \lim _{\epsilon>0} S \backslash\left(o_{\{[a-\epsilon, b+\epsilon]\}}\backslash\right) \backslash.$	$S\left(o_{[a, b]}\right) \cong \lim _{\epsilon>0} S\left(o_{[a-\epsilon, b+\epsilon]}\right) .$	$S\left(o_{[a, b]}\right) \cong \lim _{\epsilon>0} S\left(o_{[a-\epsilon, b+\epsilon]}\right) .$

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fong-spivak-seven-sketchesEQ0177	272	■ 1.000	$(h, 1), \quad (h, 2), \quad (h, 3), \quad (1, 1), \quad (1, 2), \quad (1, 3), \quad (1, 3) \quad .$	$(h, 1), \quad (h, 2), \quad (h, 3), \quad (1, 1), \quad (1, 2), \quad (1, 3).$	$(h, 1), \quad (h, 2), \quad (h, 3), \quad (1, 1), \quad (1, 2), \quad (1, 3).$
fong-spivak-seven-sketchesEQ0178	272	■ 1.000	$(h, 1), \quad (1, 1), \quad (1, 2), \quad (2, 2), \quad (3, 2) \quad .$	$(h, 1), \quad (1, 1), \quad (1, 2), \quad (2, 2), \quad (3, 2).$	$(h, 1), \quad (1, 1), \quad (1, 2), \quad (2, 2), \quad (3, 2).$
fong-spivak-seven-sketchesEQ0180	273	■ 1.000	$\begin{array}{lllll} (11, 11) & (11, 12) & (12, 11) & (12, 12) & (13, 13) \\ (12, 12) & (13, 13) & (21, 21) & (22, 22) & (12, 23) & (23, 22) & (23, 23) \end{array}$	$\begin{array}{lllll} (11, 11) & (11, 12) & (12, 11) & (12, 12) & (13, 13) \\ (21, 21) & (22, 22) & (12, 23) & (23, 22) & (23, 23) \end{array}$	$\begin{array}{lllll} (11, 11) & (11, 12) & (12, 11) & (12, 12) & (13, 13) \\ (21, 21) & (22, 22) & (12, 23) & (23, 22) & (23, 23) \end{array}$
fong-spivak-seven-sketchesEQ0182	274	■ 1.000	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4$	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4$	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4$
fong-spivak-seven-sketchesEQ0183	274	■ 1.000	$(\bullet)(\circ)(*), \quad (\bullet\circ)(*), \quad (\bullet*)(\circ), \quad (\bullet)(\circ*), \quad (\bullet\circ*)$	$(\bullet)(\circ)(*), \quad (\bullet\circ)(*), \quad (\bullet*)(\circ), \quad (\bullet)(\circ*), \quad (\bullet\circ*)$	$(\bullet)(\circ)(*), \quad (\bullet\circ)(*), \quad (\bullet*)(\circ), \quad (\bullet)(\circ*), \quad (\bullet\circ*)$
fong-spivak-seven-sketchesEQ0190	280	■ 1.000	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q) \quad .$	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q).$	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q).$
fong-spivak-seven-sketchesEQ0192	283	■ 1.000	$\begin{gathered} g(\text{false})+g(\text{false})=\infty+\infty\geq\infty=g(\text{false}\wedge\text{false}) \\ g(\text{false})+g(\text{true})=\infty+0\geq\infty=g(\text{false}\wedge\text{true}) \\ g(\text{true})+g(\text{false})=0+\infty\geq\infty=g(\text{true}\wedge\text{false}) \\ g(\text{true})+g(\text{true})=0+0\geq0=g(\text{true}\wedge\text{true}). \end{gathered}$	$\begin{gathered} g(\text{false})+g(\text{false})=\infty+\infty\geq\infty=g(\text{false}\wedge\text{false}) \\ g(\text{false})+g(\text{true})=\infty+0\geq\infty=g(\text{false}\wedge\text{true}) \\ g(\text{true})+g(\text{false})=0+\infty\geq\infty=g(\text{true}\wedge\text{false}) \\ g(\text{true})+g(\text{true})=0+0\geq0=g(\text{true}\wedge\text{true}) \end{gathered}$	$\begin{gathered} g(\text{false})+g(\text{false})=\infty+\infty\geq\infty=g(\text{false}\wedge\text{false}) \\ g(\text{false})+g(\text{true})=\infty+0\geq\infty=g(\text{false}\wedge\text{true}) \\ g(\text{true})+g(\text{false})=0+\infty\geq\infty=g(\text{true}\wedge\text{false}) \\ g(\text{true})+g(\text{true})=0+0\geq0=g(\text{true}\wedge\text{true}). \end{gathered}$
fong-spivak-seven-sketchesEQ0196	287	■ 1.000	$\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right), \quad \left(\begin{array}{ll} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array}\right), \quad \text{and} \quad \left(\begin{array}{ll} 0 & \infty \\ \infty & 0 \end{array}\right).$	$\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right), \quad \left(\begin{array}{ll} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array}\right), \quad \text{and} \quad \left(\begin{array}{ll} 0 & \infty \\ \infty & 0 \end{array}\right).$	$\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right), \quad \left(\begin{array}{ll} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array}\right), \quad \text{and} \quad \left(\begin{array}{ll} 0 & \infty \\ \infty & 0 \end{array}\right).$
fong-spivak-seven-sketchesEQ0197	287	■ 1.000	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a\right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a\right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a\right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$
fong-spivak-seven-sketchesEQ0200	288	■ 1.000	$M_X=\left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right) \quad M_X^2=\left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right) \quad M_X^3=M_X^4=\left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right)$	$M_X=\left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right) \quad M_X^2=\left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right) \quad M_X^3=M_X^4=\left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right)$	$M_X=\left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right) \quad M_X^2=\left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right) \quad M_X^3=M_X^4=\left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right)$
fong-spivak-seven-sketchesEQ0201	289	■ 1.000	$t(p):=\left(\begin{array}{l} v \text{ if } n=0 \\ t(a_n) \text{ if } n \geq 1 \end{array}\right)$	$t(p):=\left\{\begin{array}{l} v \text{ if } n=0 \\ t(a_n) \text{ if } n \geq 1 \end{array}\right.$	$t(p):=\left\{\begin{array}{l} v \text{ if } n=0 \\ t(a_n) \text{ if } n \geq 1 \end{array}\right.$
fong-spivak-seven-sketchesEQ0202	289	■ 1.000	$(p \circ q):=\left(v, a_1, \ldots, a_m, b_1, \ldots, b_n\right) \quad .$	$(p; q):=\left(v, a_1, \ldots, a_m, b_1, \ldots, b_n\right) \quad .$	$(p \circ q):=\left(v, a_1, \ldots, a_m, b_1, \ldots, b_n\right) \quad .$

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fong-spivak-seven-sketchesEQ0203	289	■ 1.000	$\left(v, a_1, \ldots, a_m, b_1, \ldots, b_n, c_1, \ldots, c_o, c_{\{1\}}, \ldots, c_{\{o\}}\right)$.	$(v, a_1, \ldots, a_m, b_1, \ldots, b_n, c_1, \ldots, c_o)$.	$(v, a_1, \ldots, a_m, b_1, \ldots, b_n, c_1, \ldots, c_o)$.
fong-spivak-seven-sketchesEQ0207	291	■ 1.000	$\begin{array}{lll} (1,1) & \& (1,2) & \& (1,3) \\ (2,1) & \& (2,2) & \& (2,3) \\ (3,1) & \& (3,2) & \& (3,3) \end{array}$	$\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}$	$\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}$
fong-spivak-seven-sketchesEQ0208	291	■ 1.000	$f^{-1}(1)=b, \quad f^{-1}(2)=a, \quad f^{-1}(3)=c$.	$f^{-1}(1) = b, \quad f^{-1}(2) = a, \quad f^{-1}(3) = c.$	$f^{-1}(1) = b, \quad f^{-1}(2) = a, \quad f^{-1}(3) = c.$
fong-spivak-seven-sketchesEQ0209	291	■ 1.000	$\begin{gathered} A^{\{\prime\}} \mapsto A, \quad f^{\{\prime\}} \mapsto f, \quad g^{\{\prime\}} \mapsto g, \quad h^{\{\prime\}} \mapsto h, \quad i^{\{\prime\}} \mapsto i, \quad B^{\{\prime\}} \mapsto B, \quad C^{\{\prime\}} \mapsto C, \quad D^{\{\prime\}} \mapsto D. \end{gathered}$	$A' \mapsto A, \quad f' \mapsto f, \quad g' \mapsto g, \quad f'; h' \mapsto f; h, \quad g'; i' \mapsto f; h, \quad B' \mapsto B, \quad h' \mapsto h, \quad C' \mapsto C, \quad i' \mapsto i, \quad D' \mapsto D.$	$A' \mapsto A, \quad f' \mapsto f, \quad g' \mapsto g, \quad f' \circ h' \mapsto f \circ h, \quad g' \circ i' \mapsto f \circ h, \quad B' \mapsto B, \quad h' \mapsto h, \quad C' \mapsto C, \quad i' \mapsto i, \quad D' \mapsto D.$
fong-spivak-seven-sketchesEQ0211	292	■ 1.000	$(F \ ; \ G)(f \ ; \ g)=G(F(f) \ ; \ F(g))=G(F(f)) \ ; \ G(F(g))$	$(F;G)(f;g) = G(F(f);F(g)) = G(F(f));G(F(g))$	$(F \circ G)(f \circ g) = G(F(f) \circ F(g)) = G(F(f)) \circ G(F(g))$
fong-spivak-seven-sketchesEQ0212	292	■ 1.000	$((F \ ; \ G) \ ; \ H)(c)=H((F \ ; \ G)(c))=H(G(F(c)))=(G \ ; \ H)(F(c))=(F \ ; \ (G \ ; \ H))(c)$.	$((F;G);H)(c) = H((F;G)(c)) = H(G(F(c))) = (G;H)(F(c)) = (F;(G;H))(c).$	$((F \circ G) \circ H)(c) = H((F \circ G)(c)) = H(G(F(c))) = (G \circ H)(F(c)) = (F \circ (G \circ H))(c).$
fong-spivak-seven-sketchesEQ0215	295	■ 1.000	$\left(f_1; f_2\right)^B(g)=g; \left(f_1; f_2\right)=\left(g; f_1\right); f_2=\left(f_1^B; f_2^B\right)(g)$	$(f_1;f_2)^B(g) = g; (f_1;f_2) = (g;f_1); f_2 = (f_1^B;f_2^B)(g)$	$(f_1 \circ f_2)^B(g) = g \circ (f_1 \circ f_2) = (g \circ f_1) \circ f_2 = (f_1^B \circ f_2^B)(g)$
fong-spivak-seven-sketchesEQ0216	296	■ 1.000	$\left(c_1, d_1\right) \rightarrow\left(c_2, d_2\right) \rightarrow\left(c_3, d_3\right) \rightarrow\left(c_4, d_4\right)$.	$(c_1, d_1) \xrightarrow{(f_1, g_1)} (c_2, d_2) \xrightarrow{(f_2, g_2)} (c_3, d_3) \xrightarrow{(f_3, g_3)} (c_4, d_4)$.	$(c_1, d_1) \xrightarrow{(f_1, g_1)} (c_2, d_2) \xrightarrow{(f_2, g_2)} (c_3, d_3) \xrightarrow{(f_3, g_3)} (c_4, d_4).$
fong-spivak-seven-sketchesEQ0220	298	■ 1.000	$\left(\mathrm{X}^{\mathrm{op}} \times \mathrm{y}\right)\left((x, y),\left(x', y'\right)\right) \leq \mathcal{V}\left(\Phi(x, y), \Phi\left(x', y'\right)\right)$.	$(\mathcal{X}^{\mathrm{op}} \times y) ((x, y), (x', y')) \leq \mathcal{V}(\Phi(x, y), \Phi(x', y'))$.	$(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y})((x, y), (x', y')) \leq \mathcal{V}(\Phi(x, y), \Phi(x', y'))$.
fong-spivak-seven-sketchesEQ0231	306	■ 1.000	$\sigma_{m,n}(i):=\begin{cases} i+n & \text{if } i \leq m \\ i-m & \text{if } m+1 \leq i \end{cases}$	$\sigma_{m,n}(i) := \begin{cases} i+n & \text{if } i \leq m \\ i-m & \text{if } m+1 \leq i \end{cases}$	$\sigma_{m,n}(i) := \begin{cases} i+n & \text{if } i \leq m \\ i-m & \text{if } m+1 \leq i \end{cases}$
fong-spivak-seven-sketchesEQ0232	306	■ 1.000	$(f+g)(i):=\begin{cases} f(i) & \text{if } i \leq m \\ g(i-m) & \text{if } m+1 \leq i \end{cases}$	$(f+g)(i) := \begin{cases} f(i) & \text{if } i \leq m \\ g(i-m) & \text{if } m+1 \leq i \end{cases}$	$(f+g)(i) := \begin{cases} f(i) & \text{if } i \leq m \\ g(i-m) & \text{if } m+1 \leq i \end{cases}$
fong-spivak-seven-sketchesEQ0233	308	■ 1.000	$a^{\{0\}}, a^{\{1\}}, a^{\{2\}}, a^{\{3\}}, \ldots, a^{\{2019\}}, \ldots$	$a^0, a^1, a^2, a^3, \ldots, a^{2019}, \ldots$	$a^0, a^1, a^2, a^3, \ldots, a^{2019}, \ldots$
fong-spivak-seven-sketchesEQ0234	308	■ 1.000	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \ldots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \ldots$	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \ldots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \ldots$	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \ldots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \ldots$
fong-spivak-seven-sketchesEQ0235	308	■ 1.000	$G:=\left\{\rho _m:n\rightarrow m\mid m,n\in \mathbb{N}\right\}, \quad s\left(\rho _m\right):=m, \quad t\left(\rho _m\right):=n;$	$G := \{\rho_{m,n} : m \rightarrow n \mid m, n \in \mathbb{N}\}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n;$	$G := \{\rho_{m,n} : m \rightarrow n \mid m, n \in \mathbb{N}\}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n;$
fong-spivak-seven-sketchesEQ0236	309	■ 1.000	$\left(\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$.	$\left(\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$.	$\left(\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches EQ0237	309	1.000	$A * B = \left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array} \right) * \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right) \neq \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right) * \left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array} \right) = B * A$	$A * B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} * \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} * \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = B * A$	$A * B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} * \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} * \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = B * A$
fong-spivak-seven-sketches EQ0238	309	1.000	$A+B=\left(\begin{array}{lllllll} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{array}\right)$	$A+B=\begin{pmatrix} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{pmatrix}$	$A+B=\begin{pmatrix} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{pmatrix}$
fong-spivak-seven-sketches EQ0239	310	1.000	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$
fong-spivak-seven-sketches EQ0240	310	1.000	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$
fong-spivak-seven-sketches EQ0241	310	1.000	$g_{\{1\}}:=c_{\{n\}}+\cdots+c_{\{n\}}: m \rightarrow(m \times n),$	$g_1:=c_n+\cdots+c_n: m \rightarrow(m \times n),$	$g_1:=c_n+\cdots+c_n: m \rightarrow(m \times n),$
fong-spivak-seven-sketches EQ0243	310	1.000	$\begin{aligned} g_{\{2\}}:= & s_{\{M(1,1)\}}+\cdots+s_{\{M(1, n)\}} \quad \& +s_{\{M(2,1)\}}+\cdots+s_{\{M(2, n)\}} \quad \& +\cdots \\ & \& +s_{\{M(m, 1)\}}+\cdots+s_{\{M(m, n)\}}:(m \times n) \\ & \rightarrow(m \times n), \end{aligned}$	$g_2:=s_{M(1,1)}+\cdots+s_{M(1,n)} \\ +s_{M(2,1)}+\cdots+s_{M(2,n)} \\ +\cdots \\ +s_{M(m,1)}+\cdots+s_{M(m,n)}:(m \times n) \rightarrow(m \times n),$	$g_2:=s_{M(1,1)}+\cdots+s_{M(1,n)} \\ +s_{M(2,1)}+\cdots+s_{M(2,n)} \\ +\cdots \\ +s_{M(m,1)}+\cdots+s_{M(m,n)}:(m \times n) \rightarrow(m \times n),$
fong-spivak-seven-sketches EQ0244	310	1.000	$g_{\{4\}}:=a_{\{m\}}+\cdots+a_{\{m\}}:(m \times n) \rightarrow n,$	$g_4:=a_m+\cdots+a_m:(m \times n) \rightarrow n,$	$g_4:=a_m+\cdots+a_m:(m \times n) \rightarrow n,$
fong-spivak-seven-sketches EQ0246	312	1.000	$\mu:=\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$	$\mu:=\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\mu:=\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$
fong-spivak-seven-sketches EQ0247	313	1.000	$U(\mu)\left(\left(a_1, \ldots, a_m\right),\left(b_1, \ldots, b_m\right)\right):=\left(a_1+b_1, \ldots, a_m+b_m\right) .$	$U(\mu)\left(\left(a_1, \ldots, a_m\right),\left(b_1, \ldots, b_m\right)\right):=\left(a_1+b_1, \ldots, a_m+b_m\right) .$	$U(\mu)\left(\left(a_1, \ldots, a_m\right),\left(b_1, \ldots, b_m\right)\right):=\left(a_1+b_1, \ldots, a_m+b_m\right) .$
fong-spivak-seven-sketches EQ0248	313	1.000	$B+C:=\left\{\left(w, y, x, z\right) \in R^{m+p} \times R^{n+q} \mid\left(w, x\right) \in B \text { and }\left(y, z\right) \in C\right\} .$	$B+C:=\left\{\left(w, y, x, z\right) \in R^{m+p} \times R^{n+q} \mid\left(w, x\right) \in B \text { and }\left(y, z\right) \in C\right\} .$	$B+C:=\left\{\left(w, y, x, z\right) \in R^{m+p} \times R^{n+q} \mid\left(w, x\right) \in B \text { and }\left(y, z\right) \in C\right\} .$
fong-spivak-seven-sketches EQ0249	313	1.000	$\begin{aligned} \mathrm{B}(g) & =\left\{\left(x, z\right) \in R^m \times R^n \mid S(g)(x)=z\right\} \\ \mathrm{B}\left(h^{\mathrm{op}}\right) & =\left\{\left(z, y\right) \in R^n \times R^\ell \mid z=S(h)(y)\right\} \end{aligned}$	$\begin{aligned} \mathrm{B}(g) & =\left\{\left(x, z\right) \in R^m \times R^n \mid S(g)(x)=z\right\} \\ \mathrm{B}\left(h^{\mathrm{op}}\right) & =\left\{\left(z, y\right) \in R^n \times R^\ell \mid z=S(h)(y)\right\} \end{aligned}$	$\begin{aligned} \mathrm{B}(g) & =\left\{\left(x, z\right) \in R^m \times R^n \mid S(g)(x)=z\right\} \\ \mathrm{B}\left(h^{\mathrm{op}}\right) & =\left\{\left(z, y\right) \in R^n \times R^\ell \mid z=S(h)(y)\right\} \end{aligned}$
fong-spivak-seven-sketches EQ0254	313	1.000	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right) ; h\right)=\left\{\left(S(g)(x), S(h)(x)\right) \mid x \in R^m\right\} .$	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right) ; h\right)=\left\{\left(S(g)(x), S(h)(x)\right) \mid x \in R^m\right\} .$	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right) ; h\right)=\left\{\left(S(g)(x), S(h)(x)\right) \mid x \in R^m\right\} .$
fong-spivak-seven-sketches EQ0255	315	1.000	$f(m)=f\left(\underbrace{1+1+\cdots+1}_m\right)=\underbrace{f(1)+f(1)+\cdots+f(1)}_m=\underbrace{1_R+1_R+\cdots+1_R}_m .$	$f(m)=f\left(\underbrace{1+1+\cdots+1}_m\right)=\underbrace{f(1)+f(1)+\cdots+f(1)}_m=\underbrace{1_R+1_R+\cdots+1_R}_m .$	$f(m)=f\left(\underbrace{1+1+\cdots+1}_m\right)=\underbrace{f(1)+f(1)+\cdots+f(1)}_m=\underbrace{1_R+1_R+\cdots+1_R}_m .$

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fong-spivak-seven-sketches EQ0261 (A.3)	327	■ 1.000	$\begin{array}{cc c c} P & Q & P \wedge Q & P = (P \wedge Q) \\ \hline \text{true} & \text{true} & \text{true} & \text{true} \\ \text{true} & \text{false} & \text{false} & \text{false} \\ \text{false} & \text{true} & \text{false} & \text{true} \\ \text{false} & \text{false} & \text{false} & \text{true} \end{array}$	$U = \bigcup_{i \in I} \{x \in \mathbb{R} \mid a_i < x < b_i\}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$
fong-spivak-seven-sketches EQ0263	333	■ 1.000	$U = \bigcup_{i \in I} \{x \in \mathbb{R} \mid a_i < x < b_i\}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$
fong-spivak-seven-sketches EQ0264	333	■ 1.000	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$